

POLYNOMIAL STABILITY OF WIND TURBINE TOWER MODELS*

MOHAMED FKIRINE[†] AND LASSI PAUNONEN[†]

Abstract. We investigate the stabilization of mathematical models describing the structural dynamics of a monopile wind turbine tower. In the fore-aft plane, we show that the system becomes polynomially stable with an energy decay rate of t^{-1} under static output feedback that relies on the velocity and/or angular velocity of the nacelle. We also prove that a tuned mass damper (TMD) in the nacelle ensures polynomial stability with the same energy decay rate, offering a viable alternative to active control. In the side-to-side plane, we analyze a model incorporating a hydraulic power transmission system and prove that feedback from the nacelle's angular velocity and the generator load torque leads to polynomial stability of the closed-loop system.

Key words. wind turbine tower, SCOLE model, polynomial stability, coupled systems, energy decay

MSC codes. 35B35, 35B40, 93C20, 93D15, 47D06, 47A10

DOI. 10.1137/25M1746847

1. Introduction. Consider a wind turbine tower anchored to the ocean floor, bearing at its peak the nacelle, which can weigh several hundred tons. Because of its mass, the nacelle interacts with the flexible tower structure, creating complex dynamic behavior that requires a detailed model for accurate analysis. Wind turbines are subject to various types of vibrations due to environmental forces such as wind and waves. Among these, fore-aft vibrations (oscillations along the wind direction) and side-side vibrations (oscillations perpendicular to the wind direction) are particularly critical in evaluating the structural dynamics of the tower. These vibrations influence the turbine's stability, performance, and lifespan.

In this paper we investigate selected linear models that describe the dynamic behavior of wind turbine towers. In particular, we prove the strong asymptotic stability of the strongly continuous semigroup associated with each model, and derive an explicit decay rate for the energy of the classical solutions as $t \rightarrow \infty$. Our results are based on derivation of resolvent estimates for the generators of the strongly continuous semigroups associated with the wind turbine tower models. This analysis can be used as a basis for control design for reducing vibrations in the mathematical models.

Zhao and Weiss [33, 34, 35] used the NASA Spacecraft Control Laboratory Experiment (SCOLE) system as an analog to model the monopile wind turbine tower dynamics in both the fore-aft and side-to-side planes. In the side-to-side plane, the SCOLE system incorporates an ordinary differential equation (ODE) to represent the dynamics of the drivetrain, which transmits mechanical power from the rotor to the generator. Originally designed to model large-scale space structures, the SCOLE system consists of a long elastic beam that is either clamped at one end and connected to a movable rigid body at the other or fully clamped at both ends, as detailed in [20, 19]. This analogy provides a valuable tool for simplifying the analysis of wind turbine tower dynamics. Stability of the model can be enhanced through force and

*Received by the editors March 31, 2025; accepted for publication (in revised form) April 4, 2026; published electronically September 1, 2026.

<https://doi.org/10.1137/25M1746847>

Funding: This work was supported by the Research Council of Finland Grant 349002.

[†]Mathematics Research Centre, Faculty of Information Technology and Communication Sciences, Tampere University, 33101 Tampere, Finland (mohamed.fkirine@tuni.fi, lassi.paunonen@tuni.fi).

torque controls. Force control can be implemented by adjusting the pitch angle to regulate aerodynamic torque and rotational speed, whereas torque control utilizes an electrically driven mass within the nacelle to indirectly influence the turbine's speed.

The mathematical model, based on the SCOPE model, for the monopile wind turbine tower Σ_{sc} in the fore-aft plane has the form

$$(1.1a) \quad \rho(x)w_{tt}(x,t) = -(EIw_{xx})_{xx}(x,t), \quad (x,t) \in (0,1) \times [0,\infty),$$

$$(1.1b) \quad w(0,t) = 0, \quad w_x(0,t) = 0,$$

$$(1.1c) \quad mw_{tt}(1,t) - (EIw_{xx})_x(1,t) = u_1(t),$$

$$(1.1d) \quad Jw_{xtt}(1,t) + EI(1)w_{xx}(1,t) = u_2(t),$$

where the subscripts t and x denote derivatives with respect to the time t and the position x . Equation (1.1a) is an Euler–Bernoulli beam equation which models the dynamics of the flexible tower whose one end is clamped to the ground due to the boundary condition (1.1b). Equations (1.1c)–(1.1d) are the Newton–Euler rigid body equations that describe how the wind turbine interacts with the nacelle. In particular, $(EIw_{xx})_x(1,t)$ and $-EI(1)w_{xx}(1,t)$ are the force and the torque exerted on the nacelle by the tower at time t . The functions u_1 and u_2 are the open-loop control inputs that are applied to the nacelle. For simplicity of exposition we shall take the tower's height to be 1. In the model $w(x,t)$ is the transverse displacement of the wind tower at position x and time t , and EI and ρ are the flexural rigidity function and the mass density function of the tower. The parameters ρ and EI are functions of the spatial variable x (but independent of t), and we assume that they are strictly positive and $\rho, EI \in C^4([0,1])$. The parameters $m > 0$ and $J > 0$ are the mass and the moment of inertia of the nacelle, respectively.

Exponential stability is typically the most desirable form of stability for controlled PDE systems. In [26, 13], it has been shown that exponential stabilization of Σ_{sc} can be achieved using higher-order output feedback, such as $w_{xxt}(1,t)$ and $w_{xxxt}(1,t)$. However, implementing such feedback is challenging in practical applications. Consequently, velocity $w_t(1,t)$ and angular velocity $w_{xt}(1,t)$ emerge as more feasible signals for stabilizing the model Σ_{sc} . Unfortunately, as demonstrated in [20, 26], the system Σ_{sc} cannot be stabilized exponentially using any feedback law of the form

$$u_i(t) = -\delta_{i1}w(1,t) - \delta_{i2}w_x(1,t) - \nu_{i1}w_t(1,t) - \nu_{i2}w_{xt}(1,t),$$

where δ_{ij} and ν_{ij} are constants for $i, j = 1, 2$. Given this limitation, aiming for strong stability becomes a more practical and achievable objective. The strong stabilization of the uniform model (i.e., where EI and ρ are constant) in one specific case was studied in [20]. The nonuniform case was addressed by Zhao and Weiss [33], who demonstrated that Σ_{sc} can be strongly stabilized with either torque or force control and by utilizing feedback from the velocity or the angular velocity of the nacelle.

As the first main result of this paper we show that the system Σ_{sc} also becomes *polynomially stable* under either torque control, force control, or a combination of the two. More precisely, we show that the energy $E_{sc}(t)$ of every classical state trajectory of the system Σ_{sc} decays to zero at the rate t^{-1} as $t \rightarrow \infty$. In addition, the energy of every mild state trajectory satisfies $E_{sc}(t) \rightarrow 0$ as $t \rightarrow \infty$. These results are presented in Theorems 3.1, 3.3, and 3.4.

As our results demonstrate, force and torque control are effective in strong and polynomial stabilization of wind turbine tower models. However, in practice they may reduce the power generation efficiency and increase the blade pitch actuator usage, which can lead to fatigue. One way to address these challenges is to use passive

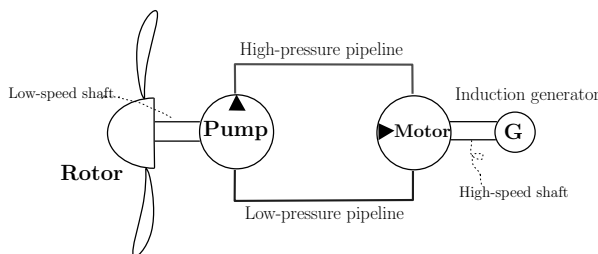


FIG. 1. Schematic of a hydraulic transmission for wind turbine.

structural control devices, the most common being the tuned mass damper (TMD). In [36, 30], the authors incorporated a TMD control system into the wind turbine tower system Σ_{sc} and showed that the resulting closed-loop system is strongly stable. As the second main result of this paper, presented in Theorem 3.5, we prove that the TMD stabilizes the monopile wind turbine model Σ_{sc} polynomially. More precisely, the energy of every classical solution of Σ_{TMD} decays to zero at the rate t^{-1} as $t \rightarrow \infty$.

In the side-side plane, we must consider the effect of the vibration torque transferred from the turbine blades to the nacelle through the dynamic mechanical transmission system. We study a model which incorporates the dynamics of a hydrostatic transmission system within the nacelle [25, 12]. The hydraulic pump converts mechanical rotational energy into hydraulic energy by forcing hydraulic fluid (usually oil or another fluid) through a closed-loop system at high pressure; see Figure 1. This high-pressure fluid is then transported through hydraulic lines to a hydraulic motor. At the receiving end, the hydraulic motor converts the energy stored in the pressurized fluid back into mechanical rotational energy, which drives a generator to produce electricity. We note that an alternative model for the dynamic behavior in the side-side plane involves a two-mass drive train and a gearbox [33, 35].

As our third main result we present conditions guaranteeing the strong and polynomial stability of the wind turbine tower system with the dynamics modeling the hydraulic pump. In particular, in Theorem 4.1 we show that the energy of classical solutions of the system decay at the rate t^{-1} as $t \rightarrow \infty$. This result is based on a new resolvent estimate for abstract coupled systems presented in Theorem 2.4.

The results in this paper are related to the study of polynomial stability of coupled systems of partial differential equations—especially beam equations—and ODEs. Polynomial stability of mathematical models of a beam with a tip mass have been studied in the case of Euler–Bernoulli beams in [10], different variants of the Timoshenko beam model in [28, 27, 6] and a piezoelectric beam model with magnetic effects in [2]. Moreover, the polynomial stabilization of beam models using dynamic boundary control has been investigated for a rotating body–beam system in [9] and for Rayleigh beam models in [32, 21]. In addition, polynomial stability of systems consisting of beam equations coupled with other one-dimensional partial differential equations have been studied, for instance, in [18, 22, 1]. The earlier studies, e.g., [20, 26, 13, 8, 33, 35, 36, 30] on the stability properties of the SCOLE model and its variants focus on strong and exponential stability. The focus of the stability analysis in this paper is on polynomial stability and rational energy decay rates which have not been previously investigated for the SCOLE system or wind turbine tower models.

The SCOLE model based on the Euler–Bernoulli beam is a simplified distributed parameter description of the tower nacelle interaction. More refined beam equations also incorporate additional effects such as geometric stiffness due to self-weight and

rotary inertia of infinitesimal tower segments (leading, e.g., to Rayleigh-type models) and may also account for couplings with blade and drivetrain dynamics; see the discussions in [33, 35]. Alternative distributed parameter models for wind turbine towers have been considered in the control literature, including boundary controlled nonuniform Euler–Bernoulli beam models derived via Hamilton’s principle; see [16, 15]. In the offshore context, floating-platform dynamics introduce further rigid body degrees of freedom and coupling effects; see [29, 17]. Our approach can also be employed in further research to establish strong asymptotic stability and rational energy decay rates for such more advanced wind turbine tower models and associated feedback control mechanisms.

The rest of this paper is organized as follows. Section 2 introduces the necessary background on selected classes of infinite-dimensional linear systems and polynomial stability of semigroups. In section 3 we examine the polynomial stability of the wind turbine in the fore-aft plane using feedback based on the velocity and/or angular velocity of the nacelle. In section 4 we introduce stabilization results for the hydraulic wind turbine in the side-side plane using feedback from the angular velocity of the nacelle and the angular velocity of the generator motor.

Notation. Given normed spaces X and Y , we write $\mathcal{L}(X, Y)$ for the space of bounded linear operators from X to Y . We denote the domain of a linear operator A by $D(A)$ and endow this space with the graph norm. We denote the kernel of a linear operator A by $\ker A$ and write $\sigma(A)$ and $\rho(A)$ for the spectrum and the resolvent set of A , respectively. The spectrum of A is given by $\sigma(A) = \mathbb{C} \setminus \rho(A)$, with the resolvent operator defined as $R(\lambda, A) = (\lambda - A)^{-1}$. The inner product on a Hilbert space X is denoted by $\langle \cdot, \cdot \rangle_X$. For $T \in \mathcal{L}(X)$, we define $\operatorname{Re} T = \frac{1}{2}(T + T^*)$. If p and q are two real-valued quantities, we write $p \lesssim q$ to express that $p \leq Mq$ for some constant $M > 0$ that is independent of all parameters free to vary in a given situation. For two functions $f : [0, \infty) \rightarrow [0, \infty)$ and $g : [0, \infty) \rightarrow [0, \infty)$, we write $f(t) = o(g(t))$ if $\frac{f(t)}{g(t)} \rightarrow 0$ as $t \rightarrow \infty$, and $f(t) = O(g(t))$ if $f(t) \lesssim g(t)$ as $t \rightarrow \infty$.

2. Background results on stability of abstract systems. In this section we present the abstract framework which we will use to analyze the stability of the wind turbine tower in sections 3 and 4. We also present a new result on resolvent estimates for abstract coupled systems.

Let X_1, X_2, U be Hilbert spaces. Let $L_1 : D(L_1) \subset X_1 \rightarrow X_1, G_1, K_1 : D(L_1) \subset X_1 \rightarrow U$ be linear operators. Let $A_2 : D(A_2) \subset X_2 \rightarrow X_2$ be the generator of a C_0 -semigroup on X_2 , $B_2 \in \mathcal{L}(U, X_2)$, $C_2 \in \mathcal{L}(X_2, U)$, and $D_2 \in \mathcal{L}(U)$. Consider an abstract coupled system of the form

$$(2.1) \quad \begin{cases} \dot{z}_1(t) = L_1 z_1(t), & t \geq 0, \\ \dot{z}_2(t) = A_2 z_2(t) - B_2 K_1 z_1(t), & t \geq 0, \\ G_1 z_1(t) = C_2 z_2(t) - D_2 K_1 z_1(t), & t \geq 0, \\ z_1(0) \in X_1, \quad z_2(0) \in X_2. \end{cases}$$

In our wind turbine models in section 3 the states $z_1(t)$ and $z_2(t)$ describe the dynamics of the flexible tower and the nacelle, respectively, and G_1 and K_1 are boundary trace operators which are used to describe the coupling between the two parts of the model. We may recast (2.1) as an abstract Cauchy problem

$$(2.2) \quad \begin{cases} \dot{z}(t) = Az(t), & t \geq 0, \\ z(0) = z_0, \end{cases}$$

with $z(t) = (z_1(t), z_2(t))^T$ on $X = X_1 \times X_2$, where $A : D(A) \subset X \rightarrow X$ is given by

$$(2.3a) \quad A = \begin{pmatrix} L_1 & 0 \\ -B_2 K_1 & A_2 \end{pmatrix},$$

$$(2.3b) \quad D(A) = \{(x_1, x_2)^T \in D(L_1) \times D(A_2) : (G_1 + D_2 K_1)x_1 = C_2 x_2\}.$$

When A generates a C_0 -semigroup $(\mathbb{T}(t))_{t \geq 0}$ on X and $z_0 \in X$, (2.2) has a well-defined mild solution $z \in C([0, \infty); X)$ given by the formula $z(t) = \mathbb{T}(t)z_0$, $t \geq 0$ [3, sect. 3.1]. In addition, the mild solution z is a classical solution $z \in C^1([0, \infty); X) \cap C([0, \infty); D(A))$ precisely if the initial condition satisfies $z_0 \in D(A)$. We establish the well-posedness of (2.2) under the assumptions on (G_1, L_1, K_1) and (A_2, B_2, C_2, D_2) listed in the following two definitions.

DEFINITION 2.1. *The triple (G_1, L_1, K_1) is said to be an impedance passive boundary node on (U, X_1, U) if the following hold.*

- (i) $G_1, K_1 \in \mathcal{L}(D(L_1), U)$;
- (ii) *the restriction of L_1 to $\ker G_1$ generates a C_0 -semigroup on X_1 ;*
- (iii) G_1 *has a bounded right-inverse, i.e., there exists $G_1^r \in \mathcal{L}(U, D(L_1))$ such that $G_1 G_1^r = I$;*
- (iv) $\operatorname{Re}\langle L_1 x, x \rangle_{X_1} \leq \operatorname{Re}\langle G_1 x, K_1 x \rangle_U$ *for all $x \in D(L_1)$.*

DEFINITION 2.2. *The tuple (A_2, B_2, C_2, D_2) is said to be an impedance passive linear system on (U, X_2, U) if the following hold:*

- (i) $B_2 \in \mathcal{L}(U, X_2)$, $C_2 \in \mathcal{L}(X_2, U)$, $D_2 \in \mathcal{L}(U)$;
- (ii) $A_2 : D(A_2) \subset X_2 \rightarrow X_2$ *generates a C_0 -semigroup on X_2 ;*
- (iii) $\operatorname{Re}\langle A_2 x + B_2 u, x \rangle_{X_2} \leq \operatorname{Re}\langle C_2 x + D_2 u, u \rangle_U$ *for all $x \in D(A_2)$ and $u \in U$.*

We define the *transfer function* $P_2 : \rho(A_2) \rightarrow \mathcal{L}(U)$ of the impedance passive linear system (A_2, B_2, C_2, D_2) by

$$P_2(\lambda) = C_2 R(\lambda, A_2) B_2 + D_2, \quad \lambda \in \rho(A_2).$$

The following result from [23, Thm. 3.8] shows that A in (2.3) generates a contraction semigroup and the resolvent norm $\|R(is, A)\|$ can be estimated using information on the boundary node (G_1, L_1, K_1) , the resolvent operator $\|R(is, A_2)\|$, and a lower bound for $\operatorname{Re} P_2(is)$.

THEOREM 2.3 (see [23, Thm. 3.8 and Rem. 3.10]). *Let (G_1, L_1, K_1) be an impedance passive boundary node on (U, X_1, U) , let (A_2, B_2, C_2, D_2) be an impedance passive linear system on (U, X_2, U) . Let A_0 denote the restriction of L_1 to $\ker(G_1 + K_1)$, and suppose there exists a nonempty set $E \subset \{s \in \mathbb{R} : is \in \rho(A_0) \cap \rho(A_2)\}$. Moreover, assume that there exists a function $\eta : E \rightarrow (0, \infty)$ such that $\operatorname{Re} P_2(is) \geq \eta(s)I$ for all $s \in E$. Then the operator A in (2.3) generates a contraction semigroup on X , $iE \subset \rho(A)$ and*

$$\|R(is, A)\| \lesssim \frac{(1 + \|R(is, A_0)\|)(1 + \|R(is, A_2)\|^2)}{\eta(s)}, \quad s \in E.$$

The model we study in section 4 can be represented as a coupled system consisting of two impedance passive linear systems. Let (A_1, B_1, C_1, D_1) be an impedance passive linear system on (U, X_1, U) and assume that $I + D_1 D_2$ is boundedly invertible. It is easy to check that also $I + D_2 D_1$ has a bounded inverse $I - D_2(I + D_1 D_2)^{-1} D_1$, and we can thus define $Q_1 = (I + D_1 D_2)^{-1}$ and $Q_2 = (I + D_2 D_1)^{-1}$. We consider the coupled system

$$\begin{cases} \dot{z}_1(t) = (A_1 - B_1 D_2 Q_1 C_1) z_1(t) + B_1 Q_2 C_2 z_2(t), & t \geq 0, \\ \dot{z}_2(t) = (A_2 - B_2 Q_1 D_1 C_2) z_2(t) - B_2 Q_1 C_1 z_1(t), & t \geq 0, \\ z_1(0) \in X_1, \quad z_2(0) \in X_2. \end{cases}$$

We may reformulate this coupled system as an abstract Cauchy problem (2.2) with state $z(t) = (z_1(t), z_2(t))^T$ on $X = X_1 \times X_2$, where $A: D(A) \subset X \rightarrow X$ is defined by

$$(2.4) \quad A = \begin{pmatrix} A_1 - B_1 D_2 Q_1 C_1 & B_1 Q_2 C_2 \\ -B_2 Q_1 C_1 & A_2 - B_2 Q_1 D_1 C_2 \end{pmatrix}, \quad D(A) = D(A_1) \times D(A_2).$$

If $K \in \mathcal{L}(U)$ is such that $K \geq cI$ for some $c > 0$, then Lemma 2.5 below shows that $I + D_1 K$ is boundedly invertible and the operator

$$A_K := A_1 - B_1 K (I + D_1 K)^{-1} C_1, \quad D(A_K) = D(A_1),$$

generates a contraction semigroup on X_1 . The following result shows that the norm $\|R(is, A)\|$ of the resolvent operator of A in (2.4) can be estimated using the resolvent norms $\|R(is, A_K)\|$ and $\|R(is, A_2)\|$ and a lower bound for $\operatorname{Re} P_2(is)$. This result is a variation of [23, Thm. 3.8].

THEOREM 2.4. *Let (A_1, B_1, C_1, D_1) and (A_2, B_2, C_2, D_2) be two impedance passive linear systems on the Hilbert spaces (U, X_1, U) and (U, X_2, U) , assume that $I + D_1 D_2$ is boundedly invertible, and assume that $K \in \mathcal{L}(U)$ is such that $\operatorname{Re} K \geq cI$ for some $c > 0$. Moreover, assume that there exists a set $E \subset \mathbb{R}$ and a function $\eta: E \rightarrow (0, \infty)$ such that $iE \subset \rho(A_K) \cap \rho(A_2)$ and $\operatorname{Re} P_2(is) \geq \eta(s)I$ for all $s \in E$. Then the operator A in (2.4) generates a contraction semigroup on X , $iE \subset \rho(A)$ and*

$$(2.5) \quad \|R(is, A)\| \lesssim \frac{(1 + \|R(is, A_K)\|)(1 + \|R(is, A_2)\|^2)}{\eta(s)}, \quad s \in E.$$

The proof of Theorem 2.4 is based on the following lemma.

LEMMA 2.5. *Let (A, B, C, D) be an impedance passive linear system on (U, X, U) , and let $Q \in \mathcal{L}(U)$ be such that $\operatorname{Re} Q \geq cI$ for some $c > 0$. Then $I + DQ$ and $I + QD$ are boundedly invertible. If we define*

$$\begin{aligned} A_Q &= A - BQ(I + DQ)^{-1}C, & B_Q &= B(I + QD)^{-1}, \\ C_Q &= (I + DQ)^{-1}C, & D_Q &= (I + DQ)^{-1}D, \end{aligned}$$

then (A_Q, B_Q, C_Q, D_Q) is an impedance passive linear system on (U, X, U) . Moreover, for all $\lambda \in \rho(A_Q) \cap \overline{\mathbb{C}}_+$ we have

$$\begin{aligned} \|R(is, A_Q)B_Q\|^2 &\leq c^{-1} \|R(is, A_Q)\|, \\ \|C_Q R(is, A_Q)\|^2 &\leq c^{-1} \|R(is, A_Q)\|, \\ \|C_Q R(is, A_Q)B_Q + D_Q\| &\leq c^{-1}. \end{aligned}$$

Proof. The assumption $\operatorname{Re} Q \geq cI$ implies that Q is boundedly invertible, and since $I + DQ = (Q^{-1} + D)Q$, where $\operatorname{Re}(Q^{-1} + D) \geq \operatorname{Re} Q^{-1} \geq c\|Q\|^{-2}I$ by [24, Lem. A.1(a)], we have that $I + DQ$ is boundedly invertible. Moreover, it is easy to check that $I + QD$ has a bounded inverse $I - Q(I + DQ)^{-1}D$. The operator A_Q generates a semigroup on X and B_Q, C_Q , and D_Q are bounded. To show that (A_Q, B_Q, C_Q, D_Q) satisfies condition (iii) of Definition 2.1, denote $T_1 = (I + DQ)^{-1}$, $T_2 = (I + QD)^{-1}$.

Note that $T_1 D = D T_2$ and $T_2 Q = Q T_1$. For all $x \in D(A_Q)$ and $u \in U$ the impedance passivity of (A, B, C, D) and $\operatorname{Re} Q \geq 0$ imply that

$$\begin{aligned} \operatorname{Re}\langle A_Q x + B_Q u, x \rangle &= \operatorname{Re}\langle A x + B T_2(-Q C x + u), x \rangle \\ &\leq \operatorname{Re}\langle C x + D T_2(-Q C x + u), T_2(-Q C x + u) \rangle \\ &= \operatorname{Re}\langle T_1 C x + T_1 D u, u \rangle \\ &\quad + \operatorname{Re}\langle T_1 C x + T_1 D u, -T_2 Q C x + (T_2 - I)u \rangle \\ &= \operatorname{Re}\langle C_Q x + D_Q u, u \rangle - \operatorname{Re}\langle T_1 C x + T_1 D u, Q(T_1 C x + T_1 D u) \rangle \\ &\leq \operatorname{Re}\langle C_Q x + D_Q u, u \rangle. \end{aligned}$$

This completes the proof that (A_Q, B_Q, C_Q, D_Q) is an impedance passive linear system. Now let $\lambda \in \rho(A_Q) \cap \mathbb{C}_+$ and $x \in X$. If we denote $z = R(is, A_Q)x \in D(A_Q)$, then the impedance passivity of (A, B, C, D) implies

$$\begin{aligned} \operatorname{Re}\langle x, z \rangle &= \operatorname{Re}\langle (is - A_Q)z, z \rangle = -\operatorname{Re}\langle Az + B(-Q(I + DQ)^{-1}Cz), z \rangle \\ &\geq -\operatorname{Re}\langle Cz + D(-Q(I + DQ)^{-1}Cz), -Q(I + DQ)^{-1}Cz \rangle \\ &= \operatorname{Re}\langle QC_Q z, C_Q z \rangle \geq c\|C_Q z\|^2. \end{aligned}$$

Since $C_Q z = C_Q R(is, A_Q)x$ and $\operatorname{Re}\langle x, z \rangle \leq \|x\|\|z\| = \|x\|\|R(is, A_Q)x\|$, the proof of the second estimate $\|C_Q R(is, A_Q)\|^2 \leq c^{-1}\|R(is, A_Q)\|^2$ is complete.

We note that (A^*, C^*, B^*, D^*) is also an impedance passive linear system and $\operatorname{Re} Q^* = \operatorname{Re} Q \geq cI$. Therefore, the first estimate follows from applying the second estimate to the system obtained from (A^*, C^*, B^*, D^*) with feedback $u = -Q^*y$. Finally, to prove the last estimate we let $\lambda \in \mathbb{C}_+$ and $u \in U$ and denote $P_Q(\lambda) = C_Q R(is, A_Q)B_Q + D_Q$ and $P(\lambda) = CR(\lambda, A)B + D$. Since $\operatorname{Re} Q \geq cI$ and $\operatorname{Re} P(\lambda) \geq 0$, the argument in the beginning of the proof shows that $I + QP(\lambda)$ is boundedly invertible and the feedback structure of (A_Q, B_Q, C_Q, D_Q) implies $P_Q(\lambda) = P(\lambda)(I + QP(\lambda))^{-1}$. Using this and $\operatorname{Re} P(\lambda) \geq 0$ we can estimate

$$\begin{aligned} c\|P_Q(\lambda)u\|^2 &\leq \operatorname{Re}\langle QP_Q(\lambda)u, P_Q(\lambda)u \rangle = \operatorname{Re}\langle QP(\lambda)(I + QP(\lambda))^{-1}u, P_Q(\lambda)u \rangle \\ &= \operatorname{Re}\langle u, P_Q(\lambda)u \rangle - \operatorname{Re}\langle (I + QP(\lambda))^{-1}u, P(\lambda)(I + QP(\lambda))^{-1}u \rangle \\ &\leq \|P_Q(\lambda)u\|\|u\|. \end{aligned}$$

Thus $\|P_Q(\lambda)\| \leq c^{-1}$ for all $\lambda \in \mathbb{C}_+$ and, by continuity, this estimate also holds for $\lambda \in \rho(A_Q) \cap \mathbb{C}_+$. $\square$

Proof of Theorem 2.4. Let E and η be as in the claim and fix $s \in E$. Since $\operatorname{Re} P_2(is) \geq \eta(s)I$ with $\eta(s) > 0$ by assumption, applying Lemma 2.5 to the system (A_1, B_1, C_1, D_1) with $Q = P_2(is)$ shows that $I + D_1 P_2(is)$ and $I + P_2(is)D_1$ are boundedly invertible. We define

$$\begin{aligned} A_{is} &= A_1 - B_1 P_2(is)(I + D_1 P_2(is))^{-1}C_1, \quad D(A_{is}) = D(A_1), \\ B_{is} &= B_1(I + P_2(is)D_1)^{-1}, \quad C_{is} = (I + D_1 P_2(is))^{-1}C_1, \quad D_{is} = (I + D_1 P_2(is))^{-1}D_1 \end{aligned}$$

and $P_{is}(is) = C_{is}R(is, A_{is})B_{is} + D_{is}$. Lemma 2.5 shows that $(A_{is}, B_{is}, C_{is}, D_{is})$ is an impedance passive linear system, and if $is \in \rho(A_{is})$, then

$$(2.6a) \quad \|R(is, A_{is})B_{is}\|^2 \leq \eta(s)^{-1}\|R(is, A_{is})\|,$$

$$(2.6b) \quad \|C_{is}R(is, A_{is})\|^2 \leq \eta(s)^{-1}\|R(is, A_{is})\|,$$

$$(2.6c) \quad \|P_{is}(is)\| \leq \eta(s)^{-1}.$$

Since A_{is} is a generator of a strongly continuous contraction semigroup, we have from [3, Prop. 4.3.1 and Cor. 4.3.5] that $is - A_{is}$ is boundedly invertible if and only if it is lower bounded, i.e., $\|(is - A_{is})x\| \geq c_s \|x\|$ for some $c_s > 0$ and for all $x \in D(A_{is})$. We will now show that if $is - A_{is}$ is boundedly invertible, then $is \in \rho(A)$. If we denote $A_2^{cl} = A_2 - B_2(I + D_1 D_2)^{-1} D_1 C_2$, then it is straightforward to verify that

$$is - A = \begin{pmatrix} I & -B_{is}C_2R(is, A_2) \\ 0 & I \end{pmatrix} \begin{pmatrix} is - A_{is} & 0 \\ 0 & is - A_2^{cl} \end{pmatrix} \begin{pmatrix} I & 0 \\ R(is, A_2)B_2C_{is} & I \end{pmatrix}.$$

We also note that since $is \in \rho(A_2)$ and since $I + D_1 P_2(is)$ is invertible, a direct computation shows that $R(is, A_2) - R(is, A_2)B_2D_{is}C_2R(is, A_2)$ is a bounded inverse of $is - A_2^{cl}$, and thus $is \in \rho(A_2^{cl})$. These properties imply that if $is \in \rho(A_{is})$, then $is - A$ has a bounded inverse given by

$$\begin{pmatrix} R(is, A_{is}) & R(is, A_{is})B_{is}C_2R(is, A_2) \\ -R(is, A_2)B_2C_{is}R(is, A_{is}) & R(is, A_2) - R(is, A_2)B_2P_{is}(is)C_2R(is, A_2) \end{pmatrix}.$$

Noting that $\eta(s) \leq \|P_2(is)\| \lesssim 1 + \|R(is, A_2)\|$, the estimates (2.6) imply that

$$(2.7) \quad \|R(is, A)\| \lesssim \|R(is, A_{is})\| + \frac{\|R(is, A_2)\| \|R(is, A_{is})\|^{1/2}}{\eta(s)^{1/2}} + \frac{1 + \|R(is, A_2)\|^2}{\eta(s)}.$$

Here and in the remaining part of the proof, the implicit constants in the estimates are independent of the choice of $s \in E$.

We will now show that $is - A_{is}$ is lower bounded and derive an estimate for $\|R(is, A_{is})\|$. To this end, let $x \in D(A_{is})$ and denote $y = (is - A_{is})x$. Using $A_K = A_1 - B_1K(I + D_1K)^{-1}C_1$ and denoting $B_K = B_1(I + KD_1)^{-1}$ we can write

$$\begin{aligned} y &= (is - A_1)x + B_1P_2(is)(I + D_1P_2(is))^{-1}C_1x \\ &= (is - A_K)x + B_K[P_2(is) - K](I + D_1P_2(is))^{-1}C_1x. \end{aligned}$$

Since $s \in E$, we have $is \in \rho(A_K)$ by assumption, and thus

$$(2.8) \quad x = R(is, A_K)y - R(is, A_K)B_K[P_2(is) - K]C_{is}x.$$

The impedance passivity of (A_1, B_1, C_1, D_1) and $\operatorname{Re} P_2(is) \geq \eta(s)I$ imply that

$$\begin{aligned} \operatorname{Re}\langle y, x \rangle &= \operatorname{Re}\langle (is - A_{is})x, x \rangle = -\operatorname{Re}\langle A_1x + B_1(-P_2(is)C_{is}x), x \rangle \\ &\geq -\operatorname{Re}\langle C_1x + D_1(-P_2(is)(I + D_1P_2(is))^{-1}C_1x), -P_2(is)C_{is}x \rangle \\ &= \operatorname{Re}\langle C_{is}x, P_2(is)C_{is}x \rangle \geq \eta(s)\|C_{is}x\|^2. \end{aligned}$$

This, in particular, implies $\|C_{is}x\|^2 \leq \eta(s)^{-1}\|x\|\|y\|$. Using (2.8), the above estimates and $\|R(is, A_K)B_K\|^2 \lesssim \|R(is, A_K)\|$ from Lemma 2.5 shows that

$$\|x\| \leq \|R(is, A_K)\|\|y\| + \|R(is, A_K)\|^{1/2}(\|K\| + \|P_2(is)\|) \left(\frac{\|x\|\|y\|}{\eta(s)} \right)^{1/2}.$$

Using Young's inequality leads to the estimate

$$\|x\| \lesssim \frac{(1 + \|R(is, A_K)\|)(1 + \|P_2(is)\|^2)}{\eta(s)}\|y\|.$$

Since $x \in D(A_{is})$ was arbitrary, we have that $is - A_{is}$ is lower bounded, and therefore boundedly invertible. Moreover, its inverse satisfies

$$\|R(is, A_{is})\| \lesssim \frac{(1 + \|R(is, A_K)\|)(1 + \|P_2(is)\|^2)}{\eta(s)}.$$

Finally, using the estimate (2.7) and the fact that $\|P_2(is)\| \leq \|C_2\| \|R(is, A_2)\| \|B_2\| + \|D_2\|$ leads to the desired estimate for $\|R(is, A)\|$. $\square$

Remark 2.6. If A_1 and A_2 have compact resolvents in Theorem 2.4, then A has compact resolvent as well. This follows easily from the fact that $A = A_d + \Delta$ with $A_d = \text{diag}(A_1, A_2)$ and $\Delta \in \mathcal{L}(X)$, the fact that A_d has compact resolvent, and the perturbation formula $R(\lambda, A) = R(\lambda, A_d)(I - \Delta R(\lambda, A_d))^{-1}$ for large $\lambda > 0$.

The following result can be used to verify the conditions of an impedance passive boundary node when the same property is known for a part of the node.

PROPOSITION 2.7. *Let $((\frac{G_1}{G_2}), L_1, (\frac{K_1}{K_2}))$ be an impedance passive boundary node on the spaces $(U_1 \times U_2, X_1, U_1 \times U_2)$, and let (A_2, B_2, C_2, D_2) be an impedance passive linear system on (U_2, X_2, U_2) with the transfer function $P_2 : \rho(A_2) \rightarrow \mathcal{L}(U_2)$. Assume that $\text{Re } P_2(\lambda) \geq c_\lambda I$ for some $\lambda \in \mathbb{C}_+$ and $c_\lambda > 0$ and define $X = X_1 \times X_2$, $L : D(L) \subset X \rightarrow X$, and $G, K : D(L) \subset X \rightarrow U_1$ by*

$$L \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} L_1 x_1 \\ -B_2 K_2 x_1 + A_2 x_2 \end{pmatrix}, \quad G \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = G_1 x_1, \quad K \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = K_1 x_1$$

for $(x_1, x_2)^\top \in D(L) = \{(x_1, x_2)^\top \in D(L_1) \times D(A_2) : (G_2 + D_2 K_2)x_1 = C_2 x_2\}$. Then (G_1, L, K_1) is an impedance passive boundary node on (U_1, X, U_1) .

Proof. Since $G_1, K_1 \in \mathcal{L}(D(L_1), U_1)$, it is easy to see that $G, K \in \mathcal{L}(D(L), U_1)$. For $x = (x_1, x_2)^\top \in D(L)$ we have

$$\begin{aligned} \text{Re} \langle Lx, x \rangle &= \text{Re} \langle L_1 x_1, x_1 \rangle + \text{Re} \langle A_2 x_2 + B_2(-K_2 x_1), x_2 \rangle \\ &\leq \text{Re} \langle G_1 x_1, K_1 x_1 \rangle + \text{Re} \langle G_2 x_1, K_2 x_1 \rangle + \text{Re} \langle C_2 x_2 + D_2(-K_2 x_1), -K_2 x_1 \rangle \\ &= \text{Re} \langle G_1 x_1, K_1 x_1 \rangle + \text{Re} \langle (G_2 + D_2 K_2)x_1 - C_2 x_2, K_2 x_1 \rangle \\ &= \text{Re} \langle G_1 x_1, K_1 x_1 \rangle. \end{aligned}$$

In particular, $A := L|_{\ker(G)}$ is dissipative. Since $(G_2, L_1|_{\ker(G_1)}, K_2)$ is an impedance passive boundary node, the arguments in the proof of [23, Thm. 3.8] show that $\rho(A) \cap \mathbb{C}_+ \neq \emptyset$. Thus A generates a contraction semigroup on $X = X_1 \times X_2$ by the Lumer–Phillips theorem. Therefore, (G_1, L, K_1) is an impedance passive boundary node provided that $G_1 \in \mathcal{L}(D(L), U_1)$ has a right inverse $G_1^r \in \mathcal{L}(U_1, D(L))$.

To verify the existence of G_1^r , denote $A_1 = L_1|_{\ker(G_1) \cap \ker(G_2)}$, and let $H_1 = (H_{11}, H_{12}) : \rho(A_1) \rightarrow \mathcal{L}(U_1 \times U_2, X_1)$ and $(\begin{smallmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{smallmatrix}) : \rho(A_1) \rightarrow \mathcal{L}(U_1 \times U_2)$ be the transfer functions of the boundary node $((\frac{G_1}{G_2}), L_1, (\frac{K_1}{K_2}))$ defined in [23, sect. 2]. Let $\lambda \in \mathbb{C}_+$ be such that $\text{Re } P_2(\lambda) \geq c_\lambda I$ for some $c_\lambda > 0$. Since $\text{Re } P_{22}(\lambda) \geq 0$ by [23, Prop. 2.4(b)], the argument in the beginning of the proof of Lemma 2.5 shows that $I + P_{22}(\lambda)P_2(\lambda)$ is invertible. By [23, Prop. 2.4(a)] we have $H_1(\lambda) \in \mathcal{L}(U_1 \times U_2, D(L_1))$ and

$$(2.9a) \quad \begin{pmatrix} G_1 \\ G_2 \end{pmatrix} (H_{11}(\lambda), H_{12}(\lambda)) = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix},$$

$$(2.9b) \quad \begin{pmatrix} K_1 \\ K_2 \end{pmatrix} (H_{11}(\lambda), H_{12}(\lambda)) = \begin{pmatrix} P_{11}(\lambda) & P_{12}(\lambda) \\ P_{21}(\lambda) & P_{22}(\lambda) \end{pmatrix}.$$

Define $Q : U_1 \rightarrow X$ by

$$Q = \begin{pmatrix} H_{11}(\lambda) \\ 0 \end{pmatrix} - \begin{pmatrix} H_{12}(\lambda)P_2(\lambda) \\ R(\lambda, A_2)B_2 \end{pmatrix} (I + P_{22}(\lambda)P_2(\lambda))^{-1}P_{21}(\lambda).$$

If $u \in U_1$ and $x = (x_1, x_2)^\top = Qu$, then $x \in D(L_1) \times D(A_2)$, and (2.9) imply that (omitting “ (λ) ” for brevity)

$$\begin{aligned} (G_2 + D_2K_2)x_1 - C_2x_2 &= (G_2 + D_2K_2)H_{11}u \\ &\quad + [-(G_2 + D_2K_2)H_{12}P_2 + C_2R(\lambda, A_2)B_2](I + P_{22}P_2)^{-1}P_{21}u \\ &= D_2P_{21}u + [-P_2 - D_2P_{22}P_2 + P_2 - D_2](I + P_{22}P_2)^{-1}P_{21}u \\ &= [D_2 + D_2P_{22}P_2 - D_2P_{22}P_2 - D_2](I + P_{22}P_2)^{-1}P_{21}u = 0. \end{aligned}$$

Thus the range of Q is contained in $D(L)$. Moreover, it is straightforward to show that $Q \in \mathcal{L}(U_1, D(L))$. Finally, (2.9a) implies $GQu = G_1H_{11}(\lambda)u - G_1H_{12}(\lambda)P_2(\lambda)(I + P_{22}(\lambda)P_2(\lambda))^{-1}P_{21}(\lambda)u = u$ for every $u \in U_1$. Thus $Q \in \mathcal{L}(U_1, D(L))$ is a right inverse of G and the proof is complete. $\square$

We conclude this section by recalling the following theorem from [5, Thm. 2.4]. This classical result allows us to deduce a rational decay rate for the orbits of a C_0 -semigroup based on resolvent estimates for its generator.

THEOREM 2.8. *Let $(\mathbb{T}(t))_{t \geq 0}$ be a bounded C_0 -semigroup on a Hilbert space X with generator A such that $i\mathbb{R} \subset \rho(A)$. Then for a fixed $\alpha > 0$ the following conditions are equivalent:*

- (a) $\|R(is, A)\| = O(|s|^\alpha)$ as $|s| \rightarrow \infty$,
- (b) $\|\mathbb{T}(t)x\| = o(t^{-1/\alpha})$ as $t \rightarrow \infty$ for all $x \in D(A)$.

If $A : D(A) \subset X \rightarrow X$ is the generator of a C_0 -semigroup $(\mathbb{T}(t))_{t \geq 0}$ on a Hilbert space X , then the orbits of this semigroup are the solutions of the abstract Cauchy problem (2.2). In particular, item (b) in Theorem 2.8 is equivalent to $\|z(t)\| = o(t^{-1/\alpha})$ as $t \rightarrow \infty$ for every classical solution of (2.2).

3. Polynomial stability of the wind turbine tower model in the fore-aft plane. In this section we study the stability of the wind turbine tower Σ_{sc} . We recall that the mathematical model of Σ_{sc} has the form

$$(3.1) \quad \begin{cases} \rho(x)w_{tt}(x, t) = -(EIw_{xx})_{xx}(x, t), & (x, t) \in (0, 1) \times [0, \infty), \\ w(0, t) = 0, & w_x(0, t) = 0, \\ mw_{tt}(1, t) - (EIw_{xx})_x(1, t) = u_1(t), \\ Jw_{xxt}(1, t) + EI(1)w_{xx}(1, t) = u_2(t), \end{cases}$$

where the spatially varying parameters ρ and EI belong to $C^4([0, 1])$ and are strictly positive. The constants $m > 0$ and $J > 0$ are positive. The natural state of Σ_{sc} at time t is

$$(3.2) \quad z(t) = (z_1(t), z_2(t), z_3(t), z_4(t))^\top = (w(\cdot, t), w_t(\cdot, t), w_t(1, t), w_{xt}(1, t))^\top,$$

where z_1 and z_2 are the transverse displacement and the transverse velocity of the tower, respectively, and where z_3 and z_4 are the velocity and the angular velocity of the nacelle, respectively. The energy state space of the solutions of Σ_{sc} is $X_{sc} = H_l^2(0, 1) \times L^2(0, 1) \times \mathbb{C}^2$, where

$$H_l^2(0, 1) = \{f \in H^2(0, 1) : f(0) = f'(0) = 0\}.$$

We equip the Hilbert space X_{sc} with the norm defined by

$$(3.3) \quad \|(f, g, \xi, \eta)\|_{X_{\text{sc}}}^2 = \int_0^1 [EI(x)|f''(x)|^2 + \rho(x)|g(x)|^2] dx + m|\xi|^2 + J|\eta|^2.$$

The wind turbine model Σ_{sc} can be rewritten in the form

$$(3.4a) \quad \dot{z}(t) = A_{\text{sc}}z(t) + B_{\text{sc}}v(t), \quad z(0) = z_0,$$

$$(3.4b) \quad y(t) = C_{\text{sc}}z(t)$$

on X_{sc} . Here B_{sc} is the control operator and v is the control input which is chosen based on the type of control applied to the nacelle. Similarly, the measurement y which will be used in the feedback stabilization and the corresponding C_{sc} are determined by the control scenario. The main operator $A_{\text{sc}} : D(A_{\text{sc}}) \subset X_{\text{sc}} \rightarrow X_{\text{sc}}$ is defined by

$$(3.5a) \quad A_{\text{sc}} \begin{pmatrix} f \\ g \\ \xi \\ \eta \end{pmatrix} = \begin{pmatrix} g \\ -\frac{1}{\rho}(EI f'')'' \\ \frac{1}{m}(EI f'')'(1) \\ -\frac{1}{J}(EI f'')(1) \end{pmatrix},$$

$$(3.5b) \quad D(A_{\text{sc}}) = \left\{ (f, g, \xi, \eta)^\top \in (H_l^2(0, 1) \cap H^4(0, 1)) \times H_l^2(0, 1) \times \mathbb{C}^2 : \begin{matrix} \xi = g(1) \\ \eta = g'(1) \end{matrix} \right\}.$$

In the case of each PDE system in the paper we consider mild solutions and classical solutions which are determined by the orbits of the associated strongly continuous semigroup. We also note that the squared norm $\|z(t)\|_{X_{\text{sc}}}^2$ is exactly twice the physical energy of the solution of Σ_{sc} at time t , that is, $E_{\text{sc}}(t) = \frac{1}{2}\|z(t)\|_{X_{\text{sc}}}^2$.

3.1. Combined force and torque control. In this section we consider the case where the wind turbine tower model Σ_{sc} has two control inputs: the force u_1 generated by an electrically driven mass within the nacelle, and the electric torque u_2 produced by the generator. In this setting the control operator $B_{\text{sc}} \in \mathcal{L}(\mathbb{C}^2, X_{\text{sc}})$ and the control input v are given by

$$(3.6) \quad B_{\text{sc}} = \begin{pmatrix} 0 & 0 & \frac{1}{m} & 0 \\ 0 & 0 & 0 & \frac{1}{J} \end{pmatrix}^\top, \quad v(t) = \begin{pmatrix} u_1(t) \\ u_2(t) \end{pmatrix}.$$

The output signal y in (3.4b) consists of two components, one proportional to the velocity and the second proportional to the angular velocity of the nacelle. More precisely, we let $a, b > 0$ and define the observation operator $C_{\text{sc}} \in \mathcal{L}(X_{\text{sc}}, \mathbb{C}^2)$ as

$$C_{\text{sc}} = \begin{pmatrix} 0 & 0 & a & 0 \\ 0 & 0 & 0 & b \end{pmatrix}.$$

The following theorem shows that the static feedback $v = -y$ stabilizes the system Σ_{sc} polynomially.

THEOREM 3.1. *Let $a, b > 0$. Assume that there exist $\varepsilon > 0$, $\delta > 0$, and $\zeta \in C^2([0, 1])$ such that for all $x \in [0, 1]$ we have*

$$(3.7) \quad \begin{aligned} \zeta(0) = 0, \quad 2(1 - \varepsilon)\rho(x) - (\rho\zeta)'(x) &< -\delta < 0, \\ EI(x)[(1 - \varepsilon) - 2\zeta'(x)] + \frac{1}{2}(EI\zeta)'(x) &< -\delta < 0. \end{aligned}$$

Then the wind turbine tower model Σ_{sc} described by (3.4) is stabilized polynomially by the static output feedback $v = -y$. More precisely, the energy of every mild solution of

$$(3.8) \quad \begin{cases} \rho(x)w_{tt}(x,t) = -(EIw_{xx})_{xx}(x,t), & (x,t) \in (0,1) \times [0,\infty), \\ w(0,t) = 0, \quad w_x(0,t) = 0, \\ mw_{tt}(1,t) - (EIw_{xx})_x(1,t) = -aw_t(1,t), \\ Jw_{xtt}(1,t) + EI(1)w_{xx}(1,t) = -bw_{xt}(1,t), \end{cases}$$

satisfies $E_{sc}(t) \rightarrow 0$ as $t \rightarrow \infty$, and for classical solutions we have $E_{sc}(t) = o(t^{-1})$ as $t \rightarrow \infty$.

Remark 3.2. The role of condition (3.7) is to establish a resolvent estimate for A_0 in Theorem 2.3. We note that the condition (3.7), in particular, holds when ρ and EI are constant. Indeed, in this case we can choose $\zeta(x) = 2x$.

Proof of Theorem 3.1. Let $X_1 := H_l^2(0,1) \times L^2(0,1)$ with the norm defined by $\|(f,g)^\top\|_{X_1}^2 = \int_0^1 EI(x)|f''(x)|^2 dx + \int_0^1 \rho(x)|g(x)|^2 dx$, let $U = \mathbb{C}^2$ with the Euclidean norm, and let $X_2 = \mathbb{C}^2$ with the norm defined by $\|(z_1, z_2)^\top\|_{X_2}^2 = m|z_1|^2 + J|z_2|^2$. We define the operators $L_1 : D(L_1) \subset X_1 \rightarrow X_1$ and $K_1, G_1 \in \mathcal{L}(D(L_1), U)$ by

$$(3.9) \quad L_1 \begin{pmatrix} f \\ g \end{pmatrix} = \begin{pmatrix} g \\ -\frac{1}{\rho}(EI f'')'' \end{pmatrix}, \quad K_1 \begin{pmatrix} f \\ g \end{pmatrix} = \begin{pmatrix} -(EI f'')'(1) \\ (EI f'')(1) \end{pmatrix}, \quad G_1 \begin{pmatrix} f \\ g \end{pmatrix} = \begin{pmatrix} g(1) \\ g'(1) \end{pmatrix}$$

for all $(f,g)^\top \in D(L_1) = (H^4(0,1) \cap H_l^2(0,1)) \times H_l^2(0,1)$. We define

$$A_2 = \begin{pmatrix} -\frac{a}{m} & 0 \\ 0 & -\frac{b}{J} \end{pmatrix}, \quad B_2 = \begin{pmatrix} \frac{1}{m} & 0 \\ 0 & \frac{1}{J} \end{pmatrix}, \quad C_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

and $D_2 = 0 \in \mathbb{C}^{2 \times 2}$. Then (3.8) can be recast as an abstract Cauchy problem of the form (2.2) with state $z(t) = (w(\cdot, t), w_t(\cdot, t), w_t(1, t), w_{xt}(1, t))^\top$ and with main operator A of the form (2.3). The operator G_1 is surjective and since U is finite-dimensional, G_1 is right-invertible. For all $(f,g)^\top \in D(L_1)$ integration by parts yields

$$\begin{aligned} \operatorname{Re} \left\langle L_1 \begin{pmatrix} f \\ g \end{pmatrix}, \begin{pmatrix} f \\ g \end{pmatrix} \right\rangle_{X_1} &= \operatorname{Re} \left(\langle EI g'', f'' \rangle_{L^2} - \langle (EI f'')'', g \rangle_{L^2} \right) \\ &= \operatorname{Re} \left(-(EI f'')'(1) \overline{g(1)} + EI(1) f''(1) \overline{g'(1)} \right) \\ &= \operatorname{Re} \left\langle G_1 \begin{pmatrix} f \\ g \end{pmatrix}, K_1 \begin{pmatrix} f \\ g \end{pmatrix} \right\rangle_U. \end{aligned}$$

In particular, the operator $A_1 := L_1|_{\ker(G_1)}$ is dissipative. To show that $I - A_1$ is surjective, let $(f_1, g_1)^\top \in X_1$ be arbitrary. If we define $h \in C^\infty([0,1])$ by

$$h(x) = (f_1'(1) - 2f_1(1))x^3 + (3f_1(1) - f_1'(1))x^2, \quad x \in [0,1],$$

and $f_2 = f_1 - h$ and $g_2 = g_1 - \frac{1}{\rho}(EI h'')''$, then $(f_2, g_2)^\top \in H_0^2(0,1) \times L^2(0,1)$, where $H_0^2(0,1) = \{f \in H^2(0,1) : f(0) = f'(0) = f(1) = f'(1) = 0\}$. Let $X_c := H_0^2(0,1) \times L^2(0,1)$ equipped with the norm $\|\cdot\|_{X_1}$. Similarly as in [31, sect. 6.10 and Ex. 3.4.13] (in the case of $EI \equiv 1$ and $\rho \equiv 1$), the results [31, Prop. 3.7.6 and Rem. 3.4.4] can be used to show that the operator $A_c : D(A_c) \subset X_c \rightarrow X_c$ defined by $A_c f = L_1 f$ for $f \in D(A_c) := (H^4(0,1) \cap H_0^2(0,1)) \times H_0^2(0,1)$ is skew-adjoint. In particular, there exists $(f_0, g)^\top \in D(A_c) \subset D(A_1)$ such that $(f_2, g_2)^\top = (I - A_1)(f_0, g)^\top$.

If we define $f = f_0 + h$, it is straightforward to check that $(f, g)^\top \in D(A_1)$ and that $(f_1, g_1)^\top = (I - A_1)(f, g)^\top$. Thus A_1 is maximally dissipative, and it generates a contraction semigroup on X_1 by the Lumer–Phillips theorem. These properties imply that (G_1, L_1, K_1) is an impedance passive boundary node on (U, X_1, U) .

It is easy to see that $\operatorname{Re}\langle A_2 z + B_2 u, z \rangle_{X_2} \leq \operatorname{Re}\langle C_2 z + D_2 u, u \rangle_U$ for all $z \in X_2$, $u \in U$, and therefore, (A_2, B_2, C_2, D_2) is an impedance passive linear system on (U, X_2, U) . Moreover, with the condition (3.7), it follows from [7, sect. 4.2] that the restriction A_0 of L_1 to $\ker(G_1 + K_1)$

$$A_0 = L_1, \\ D(A_0) = \{(f, g)^\top \in (H^4 \cap H_l^2) \times H_l^2 : (EIf'')'(1) = g(1), (EIf'')(1) = -g'(1)\}$$

generates an exponentially stable semigroup on X_1 . Thus we have $i\mathbb{R} \subset \rho(A_0)$ and $\sup_{s \in \mathbb{R}} \|R(is, A_0)\| < \infty$. On the other hand, a direct computation shows that the transfer function P_2 of (A_2, B_2, C_2, D_2) satisfies

$$\operatorname{Re} P_2(is) = \frac{1}{(s^2 - \frac{ab}{mJ})^2 + (\frac{b}{J} + \frac{a}{m})^2 s^2} \begin{pmatrix} \frac{a}{m} \left(s^2 + \frac{b^2}{J^2}\right) & 0 \\ 0 & \frac{b}{J} \left(s^2 + \frac{a^2}{m^2}\right) \end{pmatrix}, \quad s \in \mathbb{R}.$$

Hence there exists $M > 0$ such that $\operatorname{Re} P_2(is) \geq \frac{M}{1+s^2} I_{\mathbb{C}^2}$ for all $s \in \mathbb{R}$. Thus Theorem 2.3 with $E = \mathbb{R}$ shows that the operator A generates a contraction semigroup $(\mathbb{T}(t))_{t \geq 0}$ on $X_1 \times X_2$ and that $i\mathbb{R} \subset \rho(A)$ and $\|R(is, A)\| \lesssim 1 + s^2$ for $s \in \mathbb{R}$. The Arendt–Batty–Lyubich–Vũ theorem [3, Thm. 5.5.5(b)] implies that $\|\mathbb{T}(t)z_0\| \rightarrow 0$ as $t \rightarrow \infty$ for every $z_0 \in X$, and since the energy satisfies $E_{sc}(t) = \frac{1}{2}\|z(t)\|^2$, $t \geq 0$, we have $E_{sc}(t) \rightarrow 0$ as $t \rightarrow \infty$ for all mild solutions of (3.8). Moreover, Theorem 2.8 implies that $\|\mathbb{T}(t)z_0\| = o(t^{-1/2})$ as $t \rightarrow \infty$ for all $z_0 \in D(A)$, and therefore, $E_{sc}(t) = o(t^{-1})$ as $t \rightarrow \infty$ for classical solutions. $\square$

3.2. Torque control. In this section we consider the stabilization of the wind turbine tower model using only torque control, i.e., when $u_1 \equiv 0$ in (3.1). We use the boundary measurement $y(t) = bw_{xt}(1, t)$ with $b > 0$ in the feedback. In this case the control input is $v = u_2$ and the control operator and the observation operator in (3.5) are $B_{sc} = (0, 0, 0, \frac{1}{J})^\top$ and $C_{sc} = (0, 0, 0, b)$. Our main result is presented below.

THEOREM 3.3. *Assume that $b > 0$. The wind turbine tower model Σ_{sc} described by (3.4) is stabilized polynomially by the static torque feedback $v = -y$ from the angular velocity of the nacelle. More precisely, the energy of every mild solution of the system (3.8), with $a = 0$, satisfies $E_{sc}(t) \rightarrow 0$ as $t \rightarrow \infty$, and for classical solutions we have $E_{sc}(t) = o(t^{-1})$ as $t \rightarrow \infty$.*

Proof. We consider the space $X_1 := H_l^2(0, 1) \times L^2(0, 1) \times \mathbb{C}$ with the norm defined by $\|(f, g, h)^\top\|_{X_1}^2 = \int_0^1 EI(x)|f''(x)|^2 dx + \int_0^1 \rho(x)|g(x)|^2 dx + m|h|^2$. Define the space $U = \mathbb{C}$ with the Euclidean norm and define $X_2 = \mathbb{C}$ with the norm $\|z\|_{X_2}^2 = J|z|^2$. We define the operators $L_1 : D(L_1) \subset X_1 \rightarrow X_1$ and $K_1, G_1 \in \mathcal{L}(D(L_1), U)$ by

$$(3.10) \quad L_1 \begin{pmatrix} f \\ g \\ h \end{pmatrix} = \begin{pmatrix} g \\ -\frac{1}{\rho}(EIf'')'' \\ \frac{1}{m}(EIf'')'(1) \end{pmatrix}, \quad K_1 \begin{pmatrix} f \\ g \\ h \end{pmatrix} = (EIf'')(1), \quad G_1 \begin{pmatrix} f \\ g \\ h \end{pmatrix} = g'(1)$$

for all $(f, g, h)^\top \in D(L_1) = \{(f, g, h)^\top \in (H^4 \cap H_l^2) \times H_l^2 \times \mathbb{C} : h = g(1)\}$. Moreover, we define $A_2 = -b/J \in \mathcal{L}(X_2)$, $B_2 = 1/J \in \mathcal{L}(U, X_2)$, $C_2 = I \in \mathcal{L}(X_2, U)$, and $D_2 = 0 \in \mathcal{L}(U)$. Then (3.8) with $a = 0$ can be formulated as an abstract Cauchy

problem of the form (2.2) with state $z(t) = (w(\cdot, t), w_t(\cdot, t), w_t(1, t), w_{xt}(1, t))^T$ and with the main operator A of the form (2.3).

We can use Proposition 2.7 in showing that (G_1, L_1, K_1) is an impedance passive boundary node on (U, X_1, U) . To this end, in Proposition 2.7 we define the impedance passive boundary node $((\begin{smallmatrix} G_{11} \\ G_{12} \end{smallmatrix}), L_{11}, (\begin{smallmatrix} K_{11} \\ K_{12} \end{smallmatrix}))$ as the node “ (G_1, L_1, K_1) ” in proof of Theorem 3.1 with (3.9), but with the orders of the two inputs and the two outputs reversed. More precisely, L_{11} is the operator “ L_1 ” in (3.9) and

$$\begin{pmatrix} G_{11} \\ G_{12} \end{pmatrix} \begin{pmatrix} f \\ g \end{pmatrix} = \begin{pmatrix} g'(1) \\ g(1) \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} K_{11} \\ K_{12} \end{pmatrix} \begin{pmatrix} f \\ g \end{pmatrix} = \begin{pmatrix} (EIf'')(1) \\ -(EIf'')(1) \end{pmatrix}.$$

We define the impedance passive linear system in Proposition 2.7 as $(A_{12}, B_{12}, C_{12}, D_{12}) = (0, \frac{1}{m}, 1, 0)$ on $(\mathbb{C}, X_{12}, \mathbb{C})$ with $X_{12} = \mathbb{C}$ equipped with the norm $\|h\|_{X_{12}}^2 = m|h|^2$. Then (G_1, L_1, K_1) is exactly of the form of the node “ (G_1, L, K_1) ” in Proposition 2.7, and therefore, (G_1, L_1, K_1) is an impedance passive boundary node on (U, X_1, U) . Moreover, the restriction A_0 of L_1 to $\ker(G_1 + K_1)$

$$A_0 = L_1, \\ D(A_0) = \{(f, g, h)^\top \in (H^4 \cap H_l^2) \times H_l^2 \times \mathbb{C} : h = g(1), -(EIf'')(1) = g'(1)\}$$

generates an exponentially stable semigroup on X_1 by [10, Thm. 4.2] and thus we have $i\mathbb{R} \subset \rho(A_0)$ and $\sup_{s \in \mathbb{R}} \|R(is, A_0)\| < \infty$. It is easy to check that (A_2, B_2, C_2, D_2) is an impedance passive linear system on (U, X_2, U) and its transfer function is given by $P_2(is) = \frac{1}{is+b/J}$, and $\operatorname{Re} P_2(is) = \frac{b}{J} \left(\left(\frac{b}{J} \right)^2 + s^2 \right)^{-1}$. Thus there exists $M > 0$ such that $\operatorname{Re} P_2(is) \geq M(1 + s^2)^{-1}$ for all $s \in \mathbb{R}$. Arguing similarly as in the last paragraph of the proof of Theorem 3.1 we deduce that $E_{sc}(t) \rightarrow 0$ as $t \rightarrow \infty$ for every mild solution and $E_{sc}(t) = o(t^{-1})$ for all classical solutions. $\square$

3.3. Force control. In this section we consider the stabilization using only force control, i.e., when $u_2 \equiv 0$ in (3.1). We use the boundary measurement $y(t) = aw_t(1, t)$ with $a > 0$ in the feedback. The control input is $v = u_1$ and the control operator and the observation operator in (3.5) can be chosen as

$$B_{sc} = \begin{pmatrix} 0 & 0 & \frac{1}{m} & 0 \end{pmatrix}^\top, \quad C_{sc} = \begin{pmatrix} 0 & 0 & a & 0 \end{pmatrix}.$$

In this case we assume that Σ_{sc} is uniform, i.e., EI and ρ in (3.1) are strictly positive constants. The proof of our main result below is based on Theorem 2.3. The role of condition (3.11) is to guarantee that the resolvent of A_0 in Theorem 2.3 exists and is uniformly bounded on $i\mathbb{R}$. We note that if the condition (3.11) is not satisfied, then A_0 has at least one spectral point on $i\mathbb{R}$; see [8, Rem. 4].

THEOREM 3.4. *Assume that $a > 0$, that $EI > 0$ and $\rho > 0$ in (3.1) are constant, and that*

$$(3.11) \quad J \frac{EI}{\rho} \neq \frac{\sinh(k\pi)}{(k\pi)^3 (\cosh(k\pi) - (-1)^k)}, \quad k \in \mathbb{N}.$$

Then the wind turbine tower model Σ_{sc} described by (3.4) is stabilized polynomially by the static force feedback $v = -y$ from the velocity of the nacelle. More precisely, the energy of every mild solution of the system (3.8), with $b = 0$, satisfies $E_{sc}(t) \rightarrow 0$ as $t \rightarrow \infty$, and for classical solutions we have $E_{sc}(t) = o(t^{-1})$ as $t \rightarrow \infty$.

Proof. We consider the space $X_1 := H_l^2(0, 1) \times L^2(0, 1) \times \mathbb{C}$ equipped with the norm $\|(f, g, h)^\top\|_{X_1}^2 = EI \int_0^1 |f''(x)|^2 dx + \rho \int_0^1 |g(x)|^2 dx + J|h|^2$. We define $U = \mathbb{C}$ with the Euclidean norm and $X_2 = \mathbb{C}$ with the norm $\|z\|_{X_2}^2 = m|z|^2$. We define $L_1 : D(L_1) \subset X_1 \rightarrow X_1$ and $K_1, G_1 \in \mathcal{L}(D(L_1), U)$ by

$$L_1 \begin{pmatrix} f \\ g \\ h \end{pmatrix} = \begin{pmatrix} g \\ -\frac{EI}{\rho} f''' \\ -\frac{EI}{J} f''(1) \end{pmatrix}, \quad K_1 \begin{pmatrix} f \\ g \\ h \end{pmatrix} = -EI f'''(1), \quad G_1 \begin{pmatrix} f \\ g \\ h \end{pmatrix} = g(1)$$

for all $(f, g, h)^\top \in D(L_1) = \{(f, g, h)^\top \in (H^4 \cap H_l^2) \times H_l^2 \times \mathbb{C} : h = g'(1)\}$. Moreover, $A_2 = -a/m \in \mathcal{L}(X_2)$, $B_2 = 1/m \in \mathcal{L}(U, X_2)$, $C_2 = I \in \mathcal{L}(X_2, U)$, and $D_2 = 0 \in \mathcal{L}(U)$. Then (3.8) with $b = 0$ can be formulated as an abstract Cauchy problem of the form (2.2) with state $z(t) = (w(\cdot, t), w_t(\cdot, t), w_{xt}(1, t), w_t(1, t))^\top$ and with the main operator A of the form (2.3).

To show that (G_1, L_1, K_1) is an impedance passive boundary node on (U, X_1, U) , we define the impedance passive boundary node $((\begin{smallmatrix} G_{11} \\ G_{12} \end{smallmatrix}), L_{11}, (\begin{smallmatrix} K_{11} \\ K_{12} \end{smallmatrix}))$ in Proposition 2.7 as the node “ (G_1, L_1, K_1) ” in the proof of Theorem 3.1 with (3.9). In addition, we define the impedance passive linear system $(A_{12}, B_{12}, C_{12}, D_{12}) = (0, \frac{1}{J}, 1, 0)$ on $(\mathbb{C}, X_{12}, \mathbb{C})$ with $X_{12} = \mathbb{C}$ equipped with the norm $\|h\|_{X_{12}} = J|h|^2$. Then (G_1, L_1, K_1) is exactly of the form of the node “ (G_1, L, K_1) ” in Proposition 2.7, and therefore, (G_1, L_1, K_1) is an impedance passive boundary node. The restriction A_0 of L_1 to $\ker(G_1 + K_1)$ is

$$A_0 = L_1, \\ D(A_0) = \{(f, g, h)^\top \in (H^4 \cap H_l^2) \times H_l^2 \times \mathbb{C} : h = g'(1), EI f'''(1) = g(1)\}.$$

By [8, Thm. 4] this operator generates an exponentially stable semigroup on X_1 if (and only if) the condition (3.11) holds. Thus $i\mathbb{R} \subset \rho(A_0)$ and $\sup_{s \in \mathbb{R}} \|R(is, A_0)\| < \infty$. Moreover, (A_2, B_2, C_2, D_2) is an impedance passive linear system on (U, X_2, U) . Its transfer function is $P_2(is) = \frac{1}{is + a/m}$ and thus $\operatorname{Re} P_2(is) = \frac{a}{m} \left(\left(\frac{a}{m} \right)^2 + s^2 \right)^{-1}$ for $s \in \mathbb{R}$. Hence there exists $M > 0$ such that $\operatorname{Re} P_2(is) \geq M(1 + s^2)^{-1}$ for all $s \in \mathbb{R}$. Arguing similarly as in the last paragraph of the proof of Theorem 3.1 we deduce that $E_{sc}(t) \rightarrow 0$ as $t \rightarrow \infty$ for every mild solution and $E_{sc}(t) = o(t^{-1})$ as $t \rightarrow \infty$ for all classical solutions. $\square$

3.4. Tuned mass damper (TMD). In this section we study the stabilization of the monopile wind turbine Σ_{sc} using a tuned mass damper (TMD) system. The TMD consists of a large mass placed either on the nacelle floor using wheels or racks or suspended above the floor using cables. The mass and spring are carefully tuned to a specific frequency, allowing the TMD mass to oscillate at that frequency. The damper then reduces vibrations by dissipating the system’s energy as heat. The mathematical model Σ_{TMD} for the monopile wind turbine tower Σ_{sc} stabilized by a TMD in the fore–aft plane has the form [30, 36]

$$(3.12) \quad \begin{cases} \rho(x) w_{tt}(x, t) = -(EI w_{xx})_{xx}(x, t), & (x, t) \in (0, 1) \times [0, \infty), \\ w(0, t) = 0, \quad w_x(0, t) = 0, \\ m w_{tt}(1, t) - (EI w_{xx})_x(1, t) = k_1 [p(t) - w(1, t)] + d_1 [p_t(t) - w_t(1, t)], \\ J w_{xtt}(1, t) + EI(1) w_{xx}(1, t) = 0, \\ m_1 p_{tt}(t) = k_1 [w(1, t) - p(t)] + d_1 [w_t(1, t) - p_t(t)], \end{cases}$$

where $m_1 > 0$, $k_1 > 0$, and $d_1 > 0$ are the mass, spring constant, and damping coefficient of the TMD. Here $p(t) - w(1, t)$ describes the displacement of the mass

component of the TMD with respect to the nacelle and $p_t(t)$ describes the mass component's translational velocity with respect to the earth. The monopile wind turbine tower and the TMD subsystems are interconnected through the translational velocity of the nacelle $w_t(1, t)$ and the force $k_1(p(t) - w(1, t)) + d_1(p_t(t) - w_t(1, t))$ generated by the spring and the damper in the TMD system. The system Σ_{TMD} can be formulated as a Cauchy problem with the state

$$z(t) = \left(w(\cdot, t), w_t(\cdot, t), w_t(1, t), w_{xt}(1, t), p(t) - w(1, t), p_t(t) \right)^\top.$$

The natural energy state space of the solutions is $X = H_l^2(0, 1) \times L^2(0, 1) \times \mathbb{C}^4$ equipped with the norm

$$\|f\|_X^2 = \int_0^1 [EI(x)|f_1''(x)|^2 + \rho(x)|f_2(x)|^2] dx + m|f_3|^2 + J|f_4|^2 + k_1|f_5|^2 + m_1|f_6|^2$$

for $f = (f_1, f_2, f_3, f_4, f_5, f_6)^\top \in X$. We note that the physical energy of the solution of (3.12) satisfies $E_{\text{TMD}}(t) = \frac{1}{2}\|z(t)\|_X^2$. The operator $A : D(A) \subset X \rightarrow X$ in the abstract Cauchy problem (2.2) is defined by

$$(3.13a) \quad A \begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \\ f_6 \end{pmatrix} = \begin{pmatrix} f_2 \\ -\frac{1}{\rho}(EI f_1'')'' \\ \frac{1}{m}(EI f_1')'(1) - \frac{d_1}{m}f_3 + \frac{k_1}{m}f_5 + \frac{d_1}{m}f_6 \\ -\frac{1}{J}(EI f_1'')(1) \\ -f_3 + f_6 \\ \frac{d_1}{m_1}f_3 - \frac{k_1}{m_1}f_5 - \frac{d_1}{m_1}f_6 \end{pmatrix}$$

$$(3.13b) \quad D(A) = \{(f_1, \dots, f_6)^\top \in (H_l^2 \cap H^4) \times H_l^2 \times \mathbb{C}^4 : f_4 = f_2'(1), f_3 = f_2(1)\}.$$

Our main result below shows that if the parameters EI and ρ are constant, then the TMD stabilizes the wind turbine tower model polynomially.

THEOREM 3.5. *Assume that EI and ρ in (3.12) are strictly positive constants and satisfy*

$$(3.14) \quad J \frac{EI}{\rho} \neq \frac{\sinh(k\pi)}{(k\pi)^3 (\cosh(k\pi) - (-1)^k)}, \quad k \in \mathbb{N}.$$

Then the energy of every mild solution of the monopile wind turbine with the TMD described by (3.12) satisfies $E_{\text{TMD}}(t) \rightarrow 0$ as $t \rightarrow \infty$, and for classical solutions we have $E_{\text{TMD}}(t) = o(t^{-1})$ as $t \rightarrow \infty$.

Proof. Let $X_1 = X_{\text{sc}} = H_l^2(0, 1) \times L^2(0, 1) \times \mathbb{C}^2$ with the inner product defined by (3.3), let $U = \mathbb{C}$ with the Euclidean norm, and define $A_1 : D(A_1) \subset X_1 \rightarrow X_1$ so that $A_1 = A_{\text{sc}}$ in (3.5). Then the operator A in (3.13) has the structure

$$A = \begin{pmatrix} A_1 - B_1 D_2 C_1 & B_1 C_2 \\ -B_2 C_1 & A_2 \end{pmatrix},$$

where $B_1 = (0, 0, \frac{1}{m}, 0)^\top \in \mathcal{L}(U, X_1)$, $C_1 = (0, 0, 1, 0) \in \mathcal{L}(X_1, U)$,

$$A_2 = \begin{pmatrix} 0 & 1 \\ -\frac{k_1}{m_1} & -\frac{d_1}{m_1} \end{pmatrix}, \quad B_2 = \begin{pmatrix} 1 \\ -\frac{d_1}{m_1} \end{pmatrix}, \quad C_2 = \begin{pmatrix} k_1 & d_1 \end{pmatrix}, \quad D_2 = d_1.$$

Since the A_1 is skew-adjoint with respect to the inner product on X_1 (see [33, Prop. 3.1]) and since $C_1 = B_1^* \in \mathcal{L}(X_1, U)$, it is easy to check that (A_1, B_1, C_1, D_1) with

$D_1 = 0$ is an impedance passive linear system on (U, X_1, U) . Define the space $X_2 = \mathbb{C}^2$ with the norm $\|(z_1, z_2)^\top\|_{X_2}^2 = k_1|z_1|^2 + m_1|z_2|^2$. Then we have

$$\begin{aligned} \operatorname{Re}\langle A_2 z + B_2 u, z \rangle_{X_2} &= -d_1|z_2|^2 + k_1 \operatorname{Re}(u\bar{z}_1) - d_1 \operatorname{Re}(u\bar{z}_2), \\ \operatorname{Re}\langle C_2 z + D_2 u, u \rangle_U &= k_1 \operatorname{Re}(z_1\bar{u}) + d_1 \operatorname{Re}(z_2\bar{u}) + d_1|u|^2 \end{aligned}$$

for $z = (z_1, z_2)^\top \in X_2$ and $u \in U$. We have $\operatorname{Re}\langle A_2 z + B_2 u, z \rangle_{X_2} - \operatorname{Re}\langle C_2 z + D_2 u, u \rangle_U = -d_1|z_2 + u|^2 \leq 0$ and (A_2, B_2, C_2, D_2) is an impedance passive linear system on (U, X_2, U) . Moreover, since $\det(\lambda - A_2) = \lambda^2 + \frac{d_1}{m_1}\lambda + \frac{k_1}{m_1}$ for $\lambda \in \mathbb{C}$, the Routh–Hurwitz criterion implies that $\sigma(A_2) \subset \mathbb{C}_-$. A direct computation shows that the transfer function of (A_2, B_2, C_2, D_2) satisfies

$$\operatorname{Re} P_2(is) = \frac{d_1 m_1^2 s^4}{m(d_1^2 s^2 + (k_1 - m_1 s^2)^2)}$$

for all $s \in \mathbb{R}$. For any given $\varepsilon > 0$ there exists a constant $c_\varepsilon > 0$ such that $\operatorname{Re} P_2(is) \geq c_\varepsilon$ for all $s \in \mathbb{R} \setminus (-\varepsilon, \varepsilon)$. Theorem 3.4 shows that if $K = I$, then the operator $A_K = A_1 - B_1 K C_1$ with domain $D(A_K) = D(A_1)$ generates a contraction semigroup $\mathbb{T}_K$ on X_1 and that $\|\mathbb{T}_K z_0\|^2 = o(t^{-1})$ as $t \rightarrow \infty$ for all $z_0 \in D(A_1)$. Theorem 2.8 thus implies that $i\mathbb{R} \subset \rho(A_K)$ and $\|R(is, A_K)\| \lesssim 1 + s^2$ for $s \in \mathbb{R}$. It follows from Theorem 2.4 with $E = \{s \in \mathbb{R} : |s| > \varepsilon\}$ that A generates a contraction semigroup on X and that $\{is \in \mathbb{R} : |s| > \varepsilon\} \subset \rho(A)$ and $\|R(is, A)\| \leq M_\varepsilon(1 + s^2)$ for some $M_\varepsilon > 0$ and for $|s| > \varepsilon$. We will now show that $0 \in \rho(A)$. Since A_1 has compact resolvent by [33, Prop. 3.2] and since X_2 is finite-dimensional, Remark 2.6 implies that A has compact resolvent. It is therefore sufficient to show that $\ker A = \{0\}$. Let $f = (f_1, f_2, f_3, f_4, f_5, f_6)^\top \in D(A)$ such that $Af = 0$. Then

$$(3.15) \quad f_2 = 0, \quad f_1'''' = 0, \quad f_3 = f_2(1), \quad f_4 = f_2'(1),$$

$$(3.16) \quad -d_1 f_3 + k_1 f_5 + d_1 f_6 + EI f_1'''(1) = 0,$$

$$(3.17) \quad f_1''(1) = 0, \quad f_1(0) = f_1'(0) = f_2(0) = f_2'(0) = 0,$$

$$(3.18) \quad -f_3 + f_6 = 0, \quad d_1 f_3 - k_1 f_5 - d_1 f_6 = 0.$$

Using (3.15) we get $f_4 = f_3 = 0$ and then (3.18) implies that $f_6 = f_5 = 0$ as well. Moreover, (3.15) implies that f_1 is a polynomial of degree 3, and using (3.17) and the fact that (3.16) implies $f_1'''(1) = 0$ we get that $f_1 = 0$. Thus $0 \in \rho(A)$. Choosing $\varepsilon > 0$ above sufficiently small we deduce that $i\mathbb{R} \subset \rho(A)$ and $\|R(is, A)\| \lesssim 1 + s^2$ for $s \in \mathbb{R}$. The Arendt–Batty–Lyubich–Vũ theorem [3, Thm. 5.5.5(b)] implies that for every mild solution the energy $E_{\text{TMD}}(t) = \frac{1}{2}\|z(t)\|^2$ converges to 0 as $t \rightarrow \infty$. Moreover, $E_{\text{TMD}}(t) = o(t^{-1})$ as $t \rightarrow \infty$ for classical solutions follows from Theorem 2.8. $\square$

4. Polynomial stability of the wind turbine tower model in the side-side plane: Hydrostatic transmission. To investigate the effect of the vibration torque transferred from the turbine to the nacelle by the hydraulic pump, we study a model [25, 12] of the wind turbine tower in the plane of the turbine blades. The model Σ_{hd} consists of the nonuniform SCOLE model system coupled with a fluid power transmission model (see Figure 2) and has the form

$$(4.1a) \quad \rho(x)w_{tt}(x, t) = -(EIw_{xx})_{xx}(x, t), \quad (x, t) \in (0, 1) \times [0, \infty),$$

$$(4.1b) \quad w(0, t) = 0, \quad w_x(0, t) = 0,$$

$$(4.1c) \quad mw_{tt}(1, t) - (EIw_{xx})_x(1, t) = 0,$$

$$(4.1d) \quad \begin{aligned} Jw_{xtt}(1, t) + EI(1)w_{xx}(1, t) &= D_p P(t) + B_p \Omega_p(t) + D_m P(t) \\ &\quad - B_m \Omega_m(t) - \tau_{\text{load}}(t), \end{aligned}$$

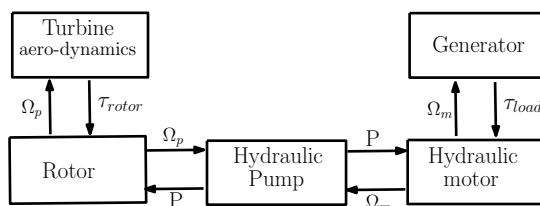


FIG. 2. Flow diagram of the wind turbine with hydrostatic transmission.

$$(4.1e) \quad \dot{\Omega}_p(t) = \frac{1}{J_T}(\tau_{\text{rotor}}(t) - B_p\Omega_p(t) - D_pP(t)) - w_{xtt}(1, t),$$

$$(4.1f) \quad \dot{\Omega}_m(t) = \frac{1}{J_G}(D_mP(t) - B_m\Omega_m(t) - \tau_{\text{load}}(t)) + w_{xtt}(1, t),$$

$$(4.1g) \quad Q_p(t) = D_p\Omega_p(t) - k_{\text{leak},p}P(t),$$

$$(4.1h) \quad Q_m(t) = D_m\Omega_m(t) + k_{\text{leak},m}P(t),$$

$$(4.1i) \quad \dot{P}(t) = \frac{\beta}{V}(Q_p(t) - Q_m(t)).$$

Equations (4.1e)–(4.1i) describe the hydrostatic transmission system. More precisely, (4.1g) is the flow equation for the pump, where Q_p is the pump flow delivery, $D_p > 0$ is the pump displacement, $k_{\text{leak},p} \geq 0$ is the pump leakage coefficient, P is the differential pressure across the pump, and Ω_p is the pump's angular velocity. Equation (4.1e) represents the torque balance between the wind turbine rotor and the pump, where the angular velocity of the nacelle, $w_{xt}(1, t)$, is also taken into consideration. In this equation, the term $J_T\Omega_p$ represents the pump's inertial torque, where $J_T > 0$ is the total inertia of the wind turbine rotor and the pump. The term $B_p\Omega_p$ represents the friction torque, where B_p is the damping coefficient of the pump, and D_pP is the pump reaction torque.

Equation (4.1h) describes the hydraulic flow supplied to the hydraulic motor, where Q_m is the motor flow delivery, $D_m > 0$ is the motor displacement, $k_{\text{leak},m} \geq 0$ is the motor leakage coefficient, P is the differential pressure across the motor, and Ω_m is the motor's angular velocity. Equation (4.1f) provides the balance between the driving torque and the braking torque for the motor shaft. The term $J_G\Omega_m$ is the motor's inertial torque, where J_G is the total inertia of the motor rotor and the generator. The term $B_m\Omega_m$ represents the motor's friction torque, where B_m is the damping coefficient of the motor, τ_{load} is the generator load torque, and D_mP is the rotor torque. Finally, (4.1i) represents the hydraulic circuit equation, where $\frac{V}{\beta}$ is the fluid capacitance of the hydraulic circuits, V is the total compressed volume from the hydraulic pump to the hydraulic motor, and β is the effective bulk modulus of the hydraulic fluid.

In our analysis it is necessary to assume that at least one of the damping coefficients, either that of the pump, B_p , or that of the motor rotor, B_m , is nonzero. We will now reformulate the system Σ_{hd} described by (4.1) as an abstract Cauchy problem of the form (2.2) with the state

$$z(t) = (w(\cdot, t), w_t(\cdot, t), w_t(1, t), w_{xt}(1, t), \Omega_p(t) + w_{xt}(1, t), \Omega_m(t) - w_{xt}(1, t), P(t))^{\top}.$$

The energy state space of the solutions of Σ_{hd} is $X_{\text{hd}} = H_l^2(0, 1) \times L^2(0, 1) \times \mathbb{C}^5$ equipped with the norm

$$\begin{aligned} \|f\|_{X_{\text{hd}}}^2 = & \int_0^1 [EI(x)|f_1''(x)|^2 + \rho(x)|f_2(x)|^2] dx + m|f_3|^2 + J|f_4|^2 + J_T|f_5|^2 \\ & + J_G|f_6|^2 + \frac{V}{\beta}|f_7|^2 \end{aligned}$$

for $f = (f_1, f_2, f_3, f_4, f_5, f_6, f_7)^\top \in X_{\text{hd}}$. We note that the physical energy of the solution of Σ_{hd} satisfies $E_{\text{hd}}(t) = \frac{1}{2}\|z(t)\|_{X_{\text{hd}}}^2$. Denoting $k_{\text{leak}} := k_{\text{leak},p} + k_{\text{leak},m}$, the main operator A_{hd} of the abstract Cauchy problem is defined by

$$A_{\text{hd}}f = \begin{pmatrix} f_2 \\ -\frac{1}{\rho}(EI f_1'')' \\ \frac{1}{m}(EI f_1'')(1) \\ -\frac{EI(1)}{J}f_1''(1) + \frac{B_p}{J}f_5 - \frac{B_p+B_m}{J}f_4 - \frac{B_m}{J}f_6 + \frac{D_p+D_m}{J}f_7 \\ -\frac{B_p}{J_T}f_5 + \frac{B_p}{J_T}f_4 - \frac{D_p}{J_T}f_7 \\ \frac{D_m}{J_G}f_7 - \frac{B_m}{J_G}f_6 - \frac{B_m}{J_G}f_4 \\ \frac{\beta}{V}(D_p f_5 - (D_p + D_m)f_4 - D_m f_6 - k_{\text{leak}}f_7) \end{pmatrix},$$

$$D(A_{\text{hd}}) = \{(f_1, \dots, f_7)^\top \in (H_l^2 \cap H^4) \times H_l^2 \times \mathbb{C}^5 : f_2(1) = f_3, f_2'(1) = f_4\}.$$

Our main result below shows that in the case where the parameters of (4.1) satisfy $J = J_G = J_T = 1$ the system Σ_{hd} without control input ($\tau_{\text{load}}(t) \equiv 0$) and without external disturbances ($\tau_{\text{rotor}}(t) \equiv 0$) is polynomially stable.

THEOREM 4.1. *Assume that $J = J_G = J_T = 1$ and $B_p + B_m > 0$. The hydraulic wind turbine model Σ_{hd} described by (4.1) is polynomially stable. More precisely, the energy of every mild solution corresponding to $\tau_{\text{rotor}}(t) \equiv 0$ and $\tau_{\text{load}}(t) \equiv 0$ satisfies $E_{\text{hd}}(t) \rightarrow 0$ as $t \rightarrow \infty$, and for classical solutions we have $E_{\text{hd}}(t) = o(t^{-1})$ as $t \rightarrow \infty$.*

The proof of Theorem 4.1 is based on Theorem 2.4 and the observation that if $J = J_G = J_T = 1$, then the operator A_{hd} has the structure

$$(4.2) \quad A_{\text{hd}} = \begin{pmatrix} A_1 - B_1 D_2 C_1 & B_1 C_2 \\ -B_2 C_1 & A_2 \end{pmatrix},$$

where $A_1 = A_{\text{sc}}$, with A_{sc} in (3.5), and $B_1 = (0, 0, 0, 1)^\top$, $C_1 = (0, 0, 0, 1)$,

$$\begin{aligned} A_2 = & \begin{pmatrix} -B_p & 0 & -D_p \\ 0 & -B_m & D_m \\ \frac{\beta}{V}D_p & -\frac{\beta}{V}D_m & -\frac{\beta}{V}k_{\text{leak}} \end{pmatrix}, \quad B_2 = \begin{pmatrix} -B_p & B_m & \frac{\beta}{V}(D_p + D_m) \end{pmatrix}^\top, \\ C_2 = & (B_p \quad -B_m \quad (D_p + D_m)), \quad D_2 = B_m + B_p. \end{aligned}$$

We define $X_2 = \mathbb{C}^3$ equipped with the norm $\|(z_1, z_2, z_3)^\top\|_{X_2}^2 = |z_1|^2 + |z_2|^2 + \frac{V}{\beta}|z_3|^2$.

Remark 4.2. Since the operator $A_1 = A_{\text{sc}}$ in (3.5) has compact resolvent by [33, Prop. 3.2], also A_{hd} has compact resolvent due to an analogous argument as in Remark 2.6. This conclusion is also valid for all $J, J_G, J_T > 0$.

The following lemma describes properties of (A_1, B_1, C_1, D_1) and (A_2, B_2, C_2, D_2) .

LEMMA 4.3. Assume that $J = J_G = J_T = 1$ and $B_p + B_m > 0$.

- (a) The tuple (A_1, B_1, C_1, D_1) with $D_1 = 0$ is an impedance passive linear system on $(\mathbb{C}, X_1, \mathbb{C})$.
- (b) The tuple (A_2, B_2, C_2, D_2) is an impedance passive linear system on $(\mathbb{C}, X_2, \mathbb{C})$ and $\sigma(A_2) \subset \mathbb{C}_-$.

Proof. Part (a) follows from the fact that A_1 is skew-adjoint by [33, Prop. 3.1] and that $C_1 = B_1^*$ and $D_1 = 0$. To prove part (b), let $z = (z_1, z_2, z_3)^T \in X_2$ and $u \in \mathbb{C}$. We have

$$\begin{aligned} \langle A_2 z + B_2 u, z \rangle_{X_2} &= -B_p |z_1|^2 - B_m |z_2|^2 - k_{\text{leak}} |z_3|^2 - B_p u \bar{z}_1 \\ &\quad + B_m u \bar{z}_2 + (D_p + D_m) u \bar{z}_3, \\ \langle C_2 z + D_2 u, u \rangle_{\mathbb{C}} &= B_p z_1 \bar{u} - B_m z_2 \bar{u} + (D_p + D_m) z_3 \bar{u} + (B_p + B_m) |u|^2 \end{aligned}$$

and thus

$$\begin{aligned} \operatorname{Re} \langle A_2 z + B_2 u, z \rangle_{X_2} - \operatorname{Re} \langle C_2 z + D_2 u, u \rangle_{\mathbb{C}} \\ = -B_p |z_1 + u|^2 - B_m |z_2 - u|^2 - k_{\text{leak}} |z_3|^2 \leq 0. \end{aligned}$$

Since X_2 is finite-dimensional, this implies that (A_2, B_2, C_2, D_2) is an impedance passive linear system on $(\mathbb{C}, X_2, \mathbb{C})$. Moreover, for $\lambda \in \mathbb{C}$ we have

$$\begin{aligned} \det(\lambda - A_2) &= \lambda^3 + (D_m^2 + D_p^2 + B_m k_{\text{leak}} + B_p k_{\text{leak}} + B_m B_p) \lambda \\ &\quad + (B_m + B_p + k_{\text{leak}}) \lambda^2 + B_p D_m^2 + B_m D_p^2 + B_m B_p k_{\text{leak}}. \end{aligned}$$

Since $B_p + B_m + k_{\text{leak}} > 0$ and $D_p, D_m > 0$ we have

$$\begin{aligned} B_p D_m^2 + B_m D_p^2 + B_m B_p k_{\text{leak}} &\leq (B_p + B_m + k_{\text{leak}}) \max\{D_m^2, D_p^2, B_m B_p\} \\ &< (B_p + B_m + k_{\text{leak}}) (D_m^2 + D_p^2 + B_m B_p) \\ &\leq (B_p + B_m + k_{\text{leak}}) (D_m^2 + D_p^2 + B_m k_{\text{leak}} + B_p k_{\text{leak}} + B_m B_p). \end{aligned}$$

Since $B_p + B_m > 0$, $D_p, D_m > 0$, and $B_p, k_{\text{leak}} \geq 0$, the above estimate and the Routh–Hurwitz criterion imply that $\det(\lambda - A_2) \neq 0$ for all $\lambda \in \mathbb{C}_+$, and thus $\sigma(A_2) \subset \mathbb{C}_-$. $\square$

Proof of Theorem 4.1. We have from Lemma 4.3 that (A_1, B_1, C_1, D_1) with $D_1 = 0$ and (A_2, B_2, C_2, D_2) are impedance passive linear systems and $\sigma(A_2) \subset \mathbb{C}_-$. To apply Theorem 2.4 we will analyze the transfer function P_2 of (A_2, B_2, C_2, D_2) . A direct computation shows that $\operatorname{Re} P_2(is) = \frac{s^2 n(s)}{d(s)}$ for all $s \in \mathbb{R} \setminus \{0\}$, where

$$\begin{aligned} n(s) &= (B_m + B_p) s^4 \\ &\quad + [(B_m B_p + k_{\text{leak}}^2)(B_m + B_p) + (D_m + D_p)^2 k] \\ &\quad + 2(B_m D_p - B_p D_m)(D_m - D_p) s^2 \\ &\quad + [(B_m D_p^2 + B_p D_m^2)(D_m - D_p)^2 + B_m B_p (B_m + B_p) k_{\text{leak}}^2 \\ &\quad + (2B_m B_p (D_m^2 + D_p^2) + (B_m D_p - B_p D_m)^2) k_{\text{leak}}], \end{aligned}$$

$$\begin{aligned} d(s) &= s^6 + [B_m^2 + B_p^2 - 2D_m^2 - 2D_p^2 + k_{\text{leak}}^2] s^4 \\ &\quad + [B_m^2 B_p^2 - 2B_m^2 D_p^2 + B_m^2 k_{\text{leak}}^2 + D_p^4 - 2B_p^2 D_m^2 + B_p^2 k_{\text{leak}}^2 + 2B_p D_p^2 k_{\text{leak}} \\ &\quad + D_m^4 + 2D_m^2 D_p^2 + 2B_m D_m^2 k_{\text{leak}}] s^2 + B_m^2 B_p^2 k_{\text{leak}}^2 + 2B_m^2 B_p D_p^2 k_{\text{leak}} \\ &\quad + B_m^2 D_p^4 + 2B_m B_p^2 D_m^2 k_{\text{leak}} + 2B_m B_p D_m^2 D_p^2 + B_p^2 D_m^4. \end{aligned}$$

Our aim is to show that $n(s) \neq 0$ for all $s \neq 0$. We have $n(s) = a_2 s^4 + a_1 s^2 + a_0$, where $a_2 = B_m + B_p > 0$ and

$$\begin{aligned} a_1 &= (B_m B_p + k_{\text{leak}}^2)(B_m + B_p) + (D_m + D_p)^2 k_{\text{leak}} + 2(B_m D_p - B_p D_m)(D_m - D_p), \\ a_0 &= (B_m D_p^2 + B_p D_m^2)(D_m - D_p)^2 + B_m B_p (B_m + B_p) k_{\text{leak}}^2 \\ &\quad + (2B_m B_p (D_m^2 + D_p^2) + (B_m D_p - B_p D_m)^2) k_{\text{leak}}. \end{aligned}$$

We first consider the case where $a_0 = 0$. Since $B_p + B_m > 0$ and $D_p, D_m > 0$, this implies that $D_m = D_p$ and further that $a_1 = (B_m B_p + k_{\text{leak}}^2)(B_m + B_p) + (D_m + D_p)^2 k_{\text{leak}} \geq 0$. Since $a_2 > 0$, $a_1 \geq 0$, and $a_0 = 0$, we have $n(s) = (a_2 s^2 + a_1) s^2 > 0$ for all $s \neq 0$ as claimed. Assume now that $a_0 > 0$. Since $a_0 a_2 > 0$, the Routh–Hurwitz criterion implies that we can have $n(s) = 0$ for some $s \neq 0$ only if $a_1 \leq 0$ and $a_1^2 - 4a_2 a_0 > 0$. Because of this, we want to show that if $a_1 \leq 0$, then necessarily $a_1^2 - 4a_2 a_0 \leq 0$. In fact, a direct computation shows that

$$\begin{aligned} a_1^2 - 4a_2 a_0 &= -2B_m B_p (B_m + B_p)(D_m + D_p)^2 k_{\text{leak}} - 2(B_m + B_p)(D_m + D_p)^2 k_{\text{leak}}^3 \\ &\quad - (B_p + B_m)^2 (B_m B_p - k_{\text{leak}}^2)^2 - (D_m + D_p)^4 k_{\text{leak}}^2 + 2a_1 (D_m + D_p)^2 k_{\text{leak}} \\ &\quad + 2a_1 (B_m B_p + k_{\text{leak}}^2)(B_m + B_p) - 4B_p B_m (D_p + D_m)^2 (D_m - D_p)^2 \\ &\quad - 4(B_p + B_m) [2B_p B_m (D_p^2 + D_m^2) + (B_m D_p - B_p D_m)^2] k_{\text{leak}}. \end{aligned}$$

This expression shows that if $a_1 \leq 0$, then $a_1^2 - 4a_2 a_0 \leq 0$. Hence $n(s) \neq 0$ for any $s > 0$. Since $\sigma(A_2) \subset \mathbb{C}_-$, we must have $d(s) \neq 0$ for $s \neq 0$. Moreover, we have $\text{Re } P_2(is) \rightarrow B_p + B_m > 0$ as $|s| \rightarrow \infty$. Combining the above properties we conclude that for any $\varepsilon > 0$ there exists $c_\varepsilon > 0$ such that $\text{Re } P_2(is) \geq c_\varepsilon I$ for all $s \in \mathbb{R} \setminus (-\varepsilon, \varepsilon)$. On the other hand, it follows from Theorem 3.3 that the operator $A_K := A_1 - B_1 K C_1$ with domain $D(A_K) = D(A_1)$ where $K = I$ generates a contraction semigroup $\mathbb{T}_K$ on X_1 and that $\|\mathbb{T}_K z_0\|^2 = o(t^{-1})$ as $t \rightarrow \infty$ for all $z_0 \in D(A_1)$. Theorem 2.8 thus implies that $i\mathbb{R} \subset \rho(A_K)$ and $\|R(is, A_K)\| \lesssim 1 + s^2$ for all $s \in \mathbb{R}$. Hence, Theorem 2.4 with $E = \{s \in \mathbb{R} : |s| > \varepsilon\}$ implies that A_{hd} generates a contraction semigroup on X_{hd} , that $\{is \in i\mathbb{R} : |s| > \varepsilon\} \subset \rho(A_{\text{hd}})$ and that for any $\varepsilon > 0$ there exists $M_\varepsilon > 0$ such that

$$(4.3) \quad \|R(is, A_{\text{hd}})\| \leq M_\varepsilon (1 + s^2), \quad \text{whenever } |s| > \varepsilon.$$

Next, we will show that $0 \in \rho(A_{\text{hd}})$. Since A_{hd} has compact resolvent by Remark 4.2. It suffices to show that $\ker A_{\text{hd}} = \{0\}$. Let $f = (f_1, f_2, f_3, f_4, f_5, f_6, f_7)^\top \in D(A_{\text{hd}})$ be such that $A_{\text{hd}} f = 0$. Then, we have

$$(4.4) \quad f_2 = 0, \quad (EI f_1'')'' = 0, \quad f_4 = f_2'(1), \quad f_3 = f_2(1),$$

$$(4.5) \quad (EI f_1'')'(1) = 0, \quad f_1(0) = f_1'(0) = 0, \quad f_2(0) = f_2'(0) = 0,$$

$$(4.6) \quad -EI(1) f_1''(1) + B_p f_5 - (B_p + B_m) f_4 - B_m f_6 + (D_p + D_m) f_7 = 0,$$

$$(4.7) \quad -B_p f_5 + B_p f_4 - D_p f_7 = 0, \quad D_m f_7 - B_m f_6 - B_m f_4 = 0,$$

$$(4.8) \quad D_p f_5 - (D_p + D_m) f_4 - D_m f_6 - k_{\text{leak}} f_7 = 0.$$

It follows from (4.4) that $f_3 = f_4 = 0$. Moreover, from (4.4), we deduce that $EI f_1''$ is an affine function. Using (4.6) and (4.7) we obtain $EI(1) f_1''(1) = 0$. Together with (4.5), this implies that $f_1'' = 0$, since EI is strictly positive. Using (4.5) we get that $f_1 = 0$. Now, using (4.7) and (4.8), with $f_4 = 0$, we get that

$$\begin{pmatrix} B_p & 0 & D_p \\ 0 & -B_m & D_m \\ D_p & -D_m & -k_{\text{leak}} \end{pmatrix} \begin{pmatrix} f_5 \\ f_6 \\ f_7 \end{pmatrix} = 0.$$

The determinant of the matrix above is $B_p B_m k_{\text{leak}} + B_p D_m^2 + B_m D_p^2$ and, since $B_p + B_m > 0$ and $D_p, D_m > 0$, it is nonsingular. Hence we have $f_5 = f_6 = f_7 = 0$. This completes the proof that $0 \in \rho(A_{\text{hd}})$. Now by choosing $\varepsilon > 0$ to be sufficiently small we can deduce that $i\mathbb{R} \subset \rho(A_{\text{hd}})$ and (4.3) implies that $\|R(is, A_{\text{hd}})\| \lesssim 1 + s^2$ for $s \in \mathbb{R}$. The Arendt–Batty–Lyubich–Vũ theorem [3, Thm. 5.5.5(b)] implies that for every mild solution the energy $E_{\text{hd}}(t) = \frac{1}{2}\|z(t)\|^2$ converges to 0 as $t \rightarrow \infty$. Finally, the energy decay rate $E_{\text{hd}}(t) = o(t^{-1})$ as $t \rightarrow \infty$ for classical solutions follows from Theorem 2.8. $\square$

We note that the general approach of applying Theorems 2.4 and 2.8 to deduce the polynomial stability of the system (4.1) is also applicable without the assumption that $J = J_G = J_T = 1$. Indeed, if the parameters of the model Σ_{hd} are fixed and if A_{hd} can be expressed in the form (4.2), then the analysis of the polynomial stability of (4.1) again reduces to showing that (A_1, B_1, C_1, D_1) and (A_2, B_2, C_2, D_2) are impedance passive linear systems and to deriving a lower bound for $\text{Re } P_2(is)$.

We conclude this section by analyzing the stability of (4.1) in the case of more general parameters J , J_G , and J_T . We show that if $J \neq J_G$, then the system can be stabilized strongly with feedback. More precisely, we assume that the external disturbance τ_{rotor} in (4.1) is zero and consider the stabilization of Σ_{hd} using the control input $u(t) = \tau_{\text{load}}(t)$. In the stabilization we consider feedback from the measurement

$$y(t) = -\frac{1}{J}w_{xt}(1, t) - \frac{1}{J_G}(\Omega_m(t) - w_{xt}(1, t)),$$

which is based on the angular velocity of the nacelle and the angular velocity of the generator motor. With these choices (4.1) can be written in the form

$$(4.9a) \quad \dot{z}(t) = A_{\text{hd}}z(t) + B_{\text{hd}}u(t),$$

$$(4.9b) \quad y(t) = B_{\text{hd}}^*z(t),$$

where $B_{\text{hd}} = (0, 0, 0, -\frac{1}{J}, 0, -\frac{1}{J_G}, 0)^\top$. The next theorem shows that this system becomes strongly stable under negative output feedback.

THEOREM 4.4. *The operator A_{hd} generates a contraction semigroup on X_{hd} . Assume that $J \neq J_G$ and $B_m > 0$. The hydraulic wind turbine model Σ_{hd} described by (4.9) is strongly stabilized by the state feedback $u = -ky$ with $k > 0$.*

Proof. We begin by proving that A_{hd} generates a contraction semigroup on X_{hd} . For any $f = (f_1, f_2, f_3, f_4, f_5, f_6, f_7)^\top \in D(A_{\text{hd}})$, we have

$$\begin{aligned} \text{Re} \langle A_{\text{hd}}f, f \rangle_{X_{\text{hd}}} &= \text{Re} \langle A_{\text{sc}}(f_1, f_2, f_3, f_4)^\top, (f_1, f_2, f_3, f_4)^\top \rangle_{X_{\text{sc}}} \\ &\quad + \text{Re} (B_p f_5 \bar{f}_4 - (B_p + B_m) f_4 \bar{f}_4 - B_m f_6 \bar{f}_4 - (D_p + D_m) f_7 \bar{f}_4) \\ &\quad + \text{Re} (-B_p f_5 \bar{f}_5 + B_p f_4 \bar{f}_5 - D_p f_7 \bar{f}_5) \\ &\quad + \text{Re} (D_m f_7 \bar{f}_6 - B_m f_6 \bar{f}_6 - B_m f_4 \bar{f}_6) \\ &\quad + \text{Re} (D_p f_5 \bar{f}_7 - (D_p + D_m) f_4 \bar{f}_7 - D_m f_6 \bar{f}_7 - k_{\text{leak}} f_7 \bar{f}_7) \\ (4.10) \quad &= -B_p |f_4 - f_5|^2 - B_m |f_6 + f_4|^2 - k_{\text{leak}} |f_7|^2 \leq 0, \end{aligned}$$

where we have used the fact that the operator A_{sc} in (3.5) is skew-adjoint by [14, Prop. 1.1]. Moreover, Remark 4.2 shows that A_{hd} has compact resolvent and, in particular, $\sigma(A_{\text{hd}}) \cap i\mathbb{R}$ is countable and A_{hd} is maximally dissipative. Thus A_{hd} generates a contraction semigroup X_{hd} by the Lumer–Phillips theorem.

The proof of the strong stability of the semigroup generated by $A_{\text{hd}} - kB_{\text{hd}}B_{\text{hd}}^*$ is based on [4, Thm. 14]. Let $\mu \in \mathbb{R}$ and $f = (f_1, \dots, f_7)^\top \in D(A_{\text{hd}})$ be such that

$A_{\text{hd}}f = i\mu f$ and $B_{\text{hd}}^*f = 0$. We will show that $f = 0$. After this, the strong stability follows immediately from [4, Thm. 14]. We have $\operatorname{Re}\langle A_{\text{hd}}f, f \rangle_{X_{\text{hd}}} = \operatorname{Re}(i\mu\|f\|_{X_{\text{hd}}}^2) = 0$ and thus (4.10) implies that $f_4 + f_6 = 0$. Since $0 = B_{\text{hd}}^*f = -\frac{f_4}{J} - \frac{f_6}{J_G}$ with $J \neq J_G$, we must have $f_4 = f_6 = 0$. The second last row of the identity $A_{\text{hd}}f = i\mu f$ implies that $f_7 = 0$, and subsequently, the last row implies $f_5 = 0$. With $f_4 = f_5 = f_6 = f_7 = 0$ the identity $A_{\text{hd}}f = i\mu f$ reduces to $A_{\text{sc}}g = i\mu g$, where $g = (f_1, f_2, f_3, 0)^\top \in D(A_{\text{sc}})$ and where A_{sc} is as in (3.5). If we define $C_{\text{sc}} = (0, 0, 0, 1)$, then $(C_{\text{sc}}, A_{\text{sc}})$ is approximately observable by [14, Cor. 2.2]. Since $C_{\text{sc}}g = 0$, the approximate observability implies that $g = 0$, which further implies $f = 0$. Since A_{hd} has compact resolvent and since μ and f were arbitrary, we have from [4, Thm. 14] that the semigroup generated by $A_{\text{hd}} - kB_{\text{hd}}B_{\text{hd}}^*$ is strongly stable. $\square$

5. Conclusions. In this paper we have analyzed the long-time behavior of several linear wind turbine tower models derived from the SCOLE framework, with an emphasis on quantitative energy decay. The stabilizing feedback mechanisms we considered (static output feedback based on the nacelle velocity and/or angular velocity, tuned mass damper, and drivetrain-type coupling) have been investigated earlier from the viewpoint of modelling and strong stabilization, and our results complement the earlier findings by establishing explicit rational decay rates for the physical energy. In the fore-aft plane, we showed that the relevant closed-loop semigroups are strongly stable and that the energy of classical solutions decays at the rate t^{-1} under torque control, force control, and combined force–torque control. We also proved that a tuned mass damper located in the nacelle yields polynomial stability with the same decay rate, providing a passive alternative to active feedback when actuator usage or power-generation efficiency is a concern. In the side-side plane, we studied a model incorporating a hydrostatic transmission subsystem and established polynomial stability of the resulting coupled system under feedback from the nacelle and generator angular velocities.

Investigating the optimality of the energy decay rates in our results is an interesting topic for future research. Using [11, Prop. 5.3] this analysis can be reduced to studying the optimality of the asymptotic resolvent growth estimates in the proofs of our main theorems. In addition, our approach can potentially be applied in the stability analysis of more elaborate wind turbine tower models, such as linearized models of floating wind turbines [17], where platform rigid-body modes and mooring-line dynamics introduce additional subsystems.

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